The utility-viz Package: Diagrams of Utility and Consumer Choice

Pin Yue Sung[*†] Ling Tak Douglas Chung[*]

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[*]Department of International Business, National Chengchi University, Taipei, Taiwan.
[†]Corresponding author. Email: contact@econ-viz.org

1 Introduction

The utility-viz package is a Python toolkit for producing publication-quality microeconomics diagrams: indifference curves, budget constraints, consumer equilibria, demand curves and exchange economies. It draws in the conventions of textbook figures—first-quadrant axes with arrow tips, LaTeX-style labels and no numeric ticks—and exports to PNG, PDF, SVG, TikZ or animated GIF.

Up to version 1.12 the package was called econ-viz. It was renamed utility-viz in 2.0.0; Section 2.8 describes what changed and how existing code keeps working. This manual describes the pre-release 2.0.0b1.

utility-viz belongs to the EconViz family of packages. It draws through mosaickit, a domain-neutral scene and rendering library, and traces curves with bezierkit. The sibling package principle-viz covers principles-of-economics diagrams.

1.1 Scope

utility-viz is organised in five parts: utility models, equilibrium solving, utility levels, drawing layers, and output formats. Models and solutions remain usable on their own, and the same model can be passed to a canvas, multi-panel figure, demand diagram, or analysis API.

1.2 Reading guide

The manual is organised in four parts (Table 1):

Table 1 Chapter guide

Topic	Subtopic	Description	Section
Drawing & export	Canvas & layers	Axes, curves, equilibria	Section 4
	Figures & demand	Panels & demand	Section 5
	Themes & config	Colours, styles, config	Section 9
	Output formats	Image files and TikZ	Section 10
Models & analysis	Built-in models	Utility functions	Section 6
	Custom & many-good	Custom, N goods, scenes	Section 7
	Comparative statics	Derivatives & Slutsky	Section 8
Animation & interaction	GIF animation	Sweeps as GIFs	Section 12
	Jupyter widgets	Sliders in notebooks	Section 13
Other inputs	Command line	Plots from the shell	Section 11
	LaTeX parser	Models from formulas	Section 14

On first use, read Section 3 and Section 4 in order. To look up a particular API, go straight to its section or to the command index. Users upgrading from econ-viz 1.x should start with Section 2.8.

2 Installation

2.1 Requirements

utility-viz requires Python 3.10 or later[1]. This chapter manages packages and projects with uv[2] and gives the matching pip command where useful.

Installing the package also installs NumPy, SciPy, matplotlib and SymPy, for array computation, numerical solving, plotting and symbolic algebra, together with mosaickit and bezierkit, for backend-neutral scenes and Bézier curves[3].

2.2 Installing uv

The commands in this manual use uv. Existing projects can keep using pip, pipx or Poetry; the package API does not depend on the tool that installed it.

Run the installer for your operating system.

macOS, Linux

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Windows

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps
1 | iex"
```

2.3 Installing utility-viz

Create a project and add utility-viz as a dependency:

```
uv init my-diagrams
cd my-diagrams
uv add utility-viz
```

Run scripts inside the project environment with `uv run`:

```
uv run python main.py
```

Save the basic example of Section 3 as `main.py` in the project directory, then run the command above. `uv add` records the dependency and `uv run` uses the project's Python environment; install and run in the same environment.

To pin the version this manual describes, give it when adding the dependency:

```
uv add "utility-viz==2.0.0b1"
```

2.0.0b1 is a pre-release. While no stable 2.x release exists, `uv add utility-viz` and `pip install utility-viz` pick it up; once a stable release is published alongside it, opt in to pre-releases with `uv add --prerelease allow utility-viz` or `pip install --pre utility-viz`.

In an existing Python virtual environment, install it with pip:

```
python -m pip install -U utility-viz
```

The package is named `utility-viz`; its Python import name is `utility_viz`:

```
from utility_viz import Canvas, solve
```

An `import` statement cannot contain a hyphen. On `ModuleNotFoundError`, check that the interpreter running the script is the one the package was installed into.

[1] The Python website provides installers for every operating system: `https://www.python.org/downloads/`.

[2] uv is a fast Python package and project manager by Astral that can also manage Python versions; see its documentation for installation and usage: `https://docs.astral.sh/uv/`.

[3] Ordinary Python plotting needs no LaTeX installation; a LaTeX distribution is only required to compile exported TikZ code.

2.4 Optional dependencies

Animation and Jupyter widgets need optional dependencies. Install the group you need:

`animation` GIF export with `Animator` (Section 12); pulls in Pillow.

`interactive` Notebook widgets with `WidgetViewer` (Section 13); pulls in ipywidgets and IPython.

`all` Every optional dependency.

```
uv add "utility-viz[animation]"    # GIF export (Pillow)
uv add "utility-viz[interactive]"  # notebook widgets
uv add "utility-viz[all]"          # all extras
```

Installing an extra does not run an animation or open a notebook by itself; call the API as shown in the relevant chapter.

2.5 Installing the command-line tool

To use only the command-line interface, install utility-viz as a standalone tool (Section 11):

```
uv tool install utility-viz
```

This installs the tool in an environment of its own. To import the package in your own Python code, still run `uv add utility-viz` in that project; calling `uv run utility-viz` inside the project keeps the CLI and your code on the same version.

2.6 Development setup

```
git clone https://github.com/EconViz/utility-viz.git
cd utility-viz
uv sync --all-packages --all-extras
```

The repository is a uv workspace with two distributions released in lockstep: utility-viz at the root and econ-viz in `packages/econ-viz/`, the compatibility package of Section 2.8. `uv sync --all-packages --all-extras` installs both in editable mode, together with the development dependencies and every optional dependency. Then run the full quality gate:

```
uv run pytest
uv run ruff check .
uv run ruff format --check .
uv run mypy .
```

2.7 Verifying the installation

```
uv run utility-viz --version   # utility-viz 2.0.0b1
uv run utility-viz help
```

The version in the comment is only an example; the output reflects the installed version. To check that Python can import the drawing and solving API:

```
uv run python -c "from utility_viz import Canvas, solve; print('OK')"
```

On a server or any environment without a display, write files with `save()` or the CLI's `--output`. `show()` needs an interactive plotting backend, so a window that fails to open does not mean the installation failed.

2.8 Migrating from econ-viz

econ-viz was renamed utility-viz in 2.0.0 (Table 2).

Table 2 Names in 1.x and 2.x

	1.x	2.x
Distribution	`pip install econ-viz`	`pip install utility-viz`
Import	`import econ_viz`	`import utility_viz`
Command	`econ-viz`	`utility-viz`

	1.x	2.x
Configuration file	`econ-viz.toml`	`utility-viz.toml`

Which package to install

utility-viz ships only the `utility_viz` package and the `utility-viz` command. econ-viz 2.x, released with the same version number, is a thin compatibility distribution: it depends on utility-viz of that version and adds the `econ_viz` package and the `econ-viz` command. Upgrading a 1.x installation with `pip install --upgrade econ-viz` (add `--pre` for a pre-release) therefore moves it onto 2.x and keeps old code running. Switch to `pip install utility-viz` once the code no longer imports `econ_viz`.

Compatibility layer

Throughout 2.x, documented 1.x code keeps working:

- `import econ_viz` emits one deprecation warning per process.
- `from econ_viz import ...` and the documented submodules (`econ_viz.models`, `econ_viz.optimizer`, `econ_viz.themes`, ...) resolve to their `utility_viz` equivalents; names that did not change are the very same objects.
- Constructing `econ_viz.Canvas`, `econ_viz.Figure` or `econ_viz.animation.Animator`, or accessing `econ_viz.Layout`, emits a `UtilityVizDeprecationWarning` naming the replacement.
- The `econ-viz` command prints a deprecation warning and forwards its arguments to `utility-viz`.
- A legacy `econ-viz.toml` is still read, with a warning (Section 9.3.3); `utility-viz init --migrate` copies it to `utility-viz.toml` (Section 11.3).

`UtilityVizDeprecationWarning`
New: v2.0.0b1

A `FutureWarning` subclass, so it is shown by default. Its message states the version that deprecated the feature, the version that removes it and the replacement, for example:

```
econ_viz.Canvas is deprecated since 2.0.0 and will be removed in 3.0.0;
use utility_viz.Canvas instead.
```

Updating code

Most code only needs the import name changed. Modules that were importable on their own in 1.x now live elsewhere (Table 3); their names are also available from the package root.

Table 3
Moved modules

1.x	2.x
`econ_viz.optimizer`	`utility_viz` (root) or `utility_viz.models`
`econ_viz.analysis`	`utility_viz.models`
`econ_viz.exceptions`	`utility_viz` (root)
`econ_viz.animation`	`utility_viz.core.animation`
`econ_viz.interactive`	`utility_viz.core.interactive`

The public API is the package root (`from utility_viz import Canvas, ...`) and `utility_viz.models`. Other `utility_viz.core.*` modules are advanced APIs that may move between minor releases.

Removal in 3.0.0

The `econ_viz` package, the `econ-viz` command and `econ-viz.toml` lookup are removed in 3.0.0, not 2.0.0. Only the documented 1.x API is covered; undocumented deep paths such as `econ_viz.canvas.renderers` resolve on a best-effort basis and may disappear at any time.

3 Quick start

3.1 A minimal example

This chapter uses the complete Cobb-Douglas consumer problem:

$$\max_{x,y} u(x,y) = x^{\frac{1}{2}} y^{\frac{1}{2}}$$
$$\text{s.t. } 2x + 3y = 30$$

The script solves the optimum and draws the indifference map, budget set, and equilibrium shown in Figure 1.

Example 1

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
lvls = levels.around(eq.utility, n=5)  # five curves and one through the optimum
cvs.add_utility(model, levels=lvls)
cvs.add_budget(2.0, 3.0, 30.0, fill=True)  # the line and its feasible set
cvs.add_equilibrium(eq, show_ray=True)  # with the expansion path
cvs.save("cobb_douglas.png")
```

Running `main.py` writes `cobb_douglas.png` to the current working directory. The optimum is $x^* = 7.5$, $y^* = 5$; spending is $2 \times 7.5 + 3 \times 5 = 30$, exactly the income. The budget line meets the axes at $\frac{I}{p_x} = 15$ and $\frac{I}{p_y} = 10$.

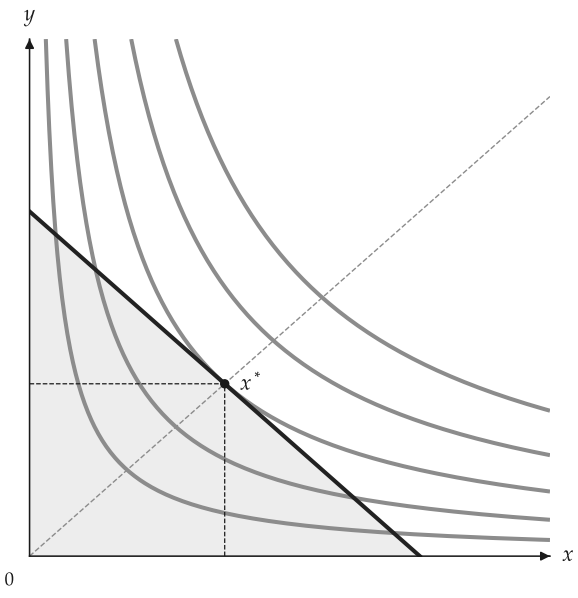


Figure 1
The consumer-equilibrium diagram.

3.2 Step by step

Every diagram follows the same steps: build a model, solve for the equilibrium, pick utility levels, create a canvas, add layers and export.

`utility_viz.models`

```
from utility_viz.models import CobbDouglas
model = CobbDouglas(alpha=0.5, beta=0.5)
```

Pick a utility function. Section 6 lists every built-in model; any Python function can also be wrapped as a model (Section 7).

`solve` `solve(⟨model⟩, px=⟨float⟩, py=⟨float⟩, income=⟨float⟩)`

Solve the consumer's problem and return an immutable `Equilibrium`.

`x` *float* Optimal quantity x^*.

`y` *float* Optimal quantity y^*.

`utility` *float* Utility at the optimum.

`bundle_type` *str* Solution type: `"interior"` for an interior solution, `"boundary"` when the numerical solution sits at a quantity lower bound (e.g. Stone-Geary subsistence), `"kink"` for a Leontief kink and `"corner"` for a perfect-substitutes corner.

```
from utility_viz import solve
eq = solve(model, px=2.0, py=3.0, income=30.0)
print(eq.x, eq.y, round(eq.utility, 3))

# 7.5 5.0 6.124
```

`levels.around` `levels.around(⟨utility⟩, n=⟨int⟩)`

Generate n utility levels centred on a given utility. Figure 1 uses `n=5`, so it draws five indifference curves.

`levels.around()` keeps the given utility among the levels, so the equilibrium lies on one of the curves. Passing `levels=5` instead lets the canvas pick levels from the sampled utility range, which need not include the equilibrium utility. To compare the same preferences across several figures, compute `lvls` once and pass it to each figure.

`Canvas` `Canvas(x_max=⟨float⟩, y_max=⟨float⟩, ...)`

Create the canvas and set the axis ranges; the other options are described in Section 4.

The axis ranges only control what is shown; they play no part in `solve()`. If the equilibrium is missing from the figure, check whether `eq.x` or `eq.y` lies outside the range and raise the axis limits.

`Canvas.add_*` Add indifference curves, a budget line or the equilibrium to the same canvas. The calls can be chained (Section 4.4). `add_equilibrium()` uses an existing solution and does not solve again; after changing prices, income or the model, call `solve()` again and build a new figure.

`Canvas.save` `Canvas.show` `save()` writes the file in the format given by its extension (Section 10); `show()` opens a matplotlib window. Output paths are relative to the working directory of the script; create any subdirectory beforehand.

Checking inputs and failed solves

Prices and income must be positive; otherwise `solve()` raises `InvalidParameterError`[4].

When the solver does not converge, `solve()` raises `OptimizationError` rather than returning an invalid equilibrium.

[4] Smooth models are solved with sequential least squares programming (SLSQP).

```python
from utility_viz import InvalidParameterError, OptimizationError, solve

try:
    eq = solve(model, px=2.0, py=3.0, income=30.0)
except (InvalidParameterError, OptimizationError) as error:
    print(error)
else:
    print(round(eq.x, 3), round(eq.y, 3), eq.bundle_type)
```

Every exception the package raises derives from `UtilityVizError`; catch it to handle all of them at once. `EconVizError`, its 1.x name, is the same class.

Numerical solutions carry small errors: a result may be close to, but not exactly, `7.5`. Compare with a tolerance when checking results, and round with `round()` only for display, so early rounding does not distort later calculations.

4 Canvas

`Canvas` builds on a matplotlib `Figure` and `Axes` and provides layers for indifference curves, budget lines, equilibria and more.

4.1 Constructor

`Canvas`

```python
from utility_viz import ArrowStyle, Canvas, Stroke, themes

cvs = Canvas(
    x_max=20,
    y_max=15,
    x_label="x",
    y_label="y",
    title=r"Cobb-Douglas $x^{0.5} y^{0.5}$",
    dpi=300,
    x_label_pos="right",   # top / right / bottom
    y_label_pos="top",     # left / top / right
    font="DejaVu Sans",
    math_font="stix",
    axis_stroke=Stroke(width=1.0, arrow=ArrowStyle.TRIANGLE),
    theme=themes.default,
)
```

Updated: v1.7.0

Create a canvas and set its axis ranges, axis labels, fonts, line styles and theme. Visual settings not passed explicitly come from the active `Config` and `Theme`.

`x_max` — *float* · default `10`
Upper bound of the horizontal axis.

`y_max` — *float* · default `10`
Upper bound of the vertical axis.

`x_label` — *str* · default `"X"`
Label at the tip of the horizontal axis.

`y_label` — *str* · default `"Y"`
Label at the tip of the vertical axis.

`title` — *str | None* · default `None`
Figure title.

`dpi` — *int* · default `300`
Raster export resolution, clamped to 1–1200.

`x_label_pos` — *str* · default `"right"`
Position of the horizontal label relative to the arrow; a `LabelPosition` is also accepted (Table 4).

Table 4 `x_label_pos` values

Value	Position
`"top"`	Above the arrow
`"right"`	Right of the arrow
`"bottom"`	Below the arrow

`y_label_pos`
str · default `"top"`

Position of the vertical label relative to the arrow (Table 5).

Table 5 `y_label_pos` values

Value	Position
`"left"`	Left of the arrow
`"top"`	Above the arrow
`"right"`	Right of the arrow

`font`
str | *list* · default `None`

Font for all text, or a list of candidate fonts. It applies to this canvas only and leaves matplotlib's global configuration alone.

`math_font`
str · default `None`

matplotlib math font set: `dejavusans`, `dejavuserif`, `cm`, `stix`, `stixsans` and so on.

`axis_stroke`
Stroke · default `theme`

Width, style, colour and arrowhead of both axes (Section 4.3).

`x_axis_stroke`
Stroke · default `None`

Override for the horizontal axis.

`y_axis_stroke`
Stroke · default `None`

Override for the vertical axis.

`theme`
Theme · default `themes.default`

Colour and style theme (Section 9).

4.2 Drawing methods

The `add_*` methods add a layer to the canvas and return the same `Canvas`, so calls can be chained (Section 4.4).

`add_utility`

```
cvs.add_utility(
    func,
    levels=3,          # count or list of levels
    color=None,        # default theme.ic_color
    linewidth=None,    # default theme.ic_linewidth
    show_rays=False,
    show_kinks=False,
    kink_radius=1.0,
    show_bliss=True,   # star at the bliss point of Satiation
    show_ic_labels=False,
    stroke=None,
    ray_stroke=None,
    ic_label=None,
    highlight_level=None,
    secondary_stroke=None,
    label_style="numeric",
)
```

Draw indifference curves for a utility model.

`levels`
int | *list* · default `3`

A number of curves, or a list of utility values such as `levels.around(eq.utility, n=5)` to place them around the optimum.

`highlight_level`
float | *None* · default `None`

Focal utility level. The closest curve is drawn in the main style.

`secondary_stroke`
Stroke | *None* · default `None`

Style of the other curves; defaults to `theme.secondary_ic_stroke`.

`show_ic_labels`
bool · default `False`

Label the curves. Labels follow the curve's tangent and stay inside the plotting area.

`label_style`
str · default `"numeric"`

`"numeric"` shows utility values; `"ordinal"` numbers the curves $u_1, u_2, \ldots$ in increasing utility.

```
lvls = levels.around(eq.utility, n=5)
cvs.add_utility(
    model,
    levels=lvls,
    highlight_level=eq.utility,
    show_ic_labels=True,
    label_style="ordinal",
)
```

Example 2

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))  # three curves around the
optimum
cvs.save("add_utility.png")
```

`add_budget`

```
cvs.add_budget(
    px, py, income,
    color=None,
    linewidth=None,
    linestyle="-",
    label=None,         # legend label in LaTeX
    fill=False,         # shade feasible set
    fill_alpha=None,    # default theme.budget_fill_alpha
    stroke=None,
)
```

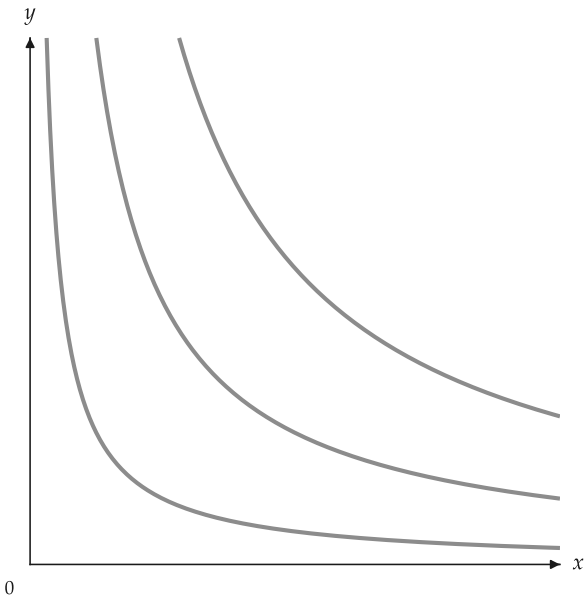


Figure 2
Three indifference curves.

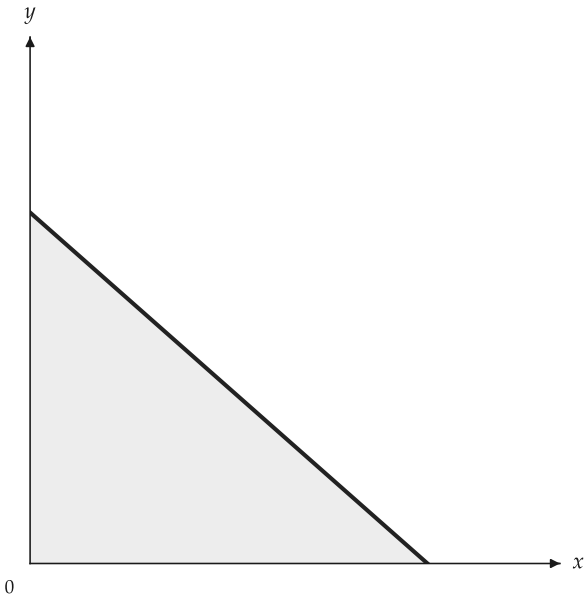


Figure 3
Budget line and shading.

Draw the budget line $p_x x + p_y y = I$.

`fill`
bool | Fill · default `False`
`True` shades the feasible set with the theme's fill; pass a `Fill` to set its colour and opacity.

`stroke`
Stroke | None · default `None`
Colour, width, style and opacity of the line in one object; takes precedence over `color`, `linewidth` and `linestyle` (Section 4.3).

`label`
str | None · default `None`
Legend text, shown once `show_legend()` is called.

Example 3

```
from utility_viz import Canvas, Config

Config.load("manual.toml").use()  # the settings behind this manual's figures

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
cvs.add_budget(2.0, 3.0, 30.0, fill=True)  # the line and its feasible set
cvs.save("add_budget.png")
```

`add_equilibrium`

```
cvs.add_equilibrium(
    eq,                 # result of solve()
    color=None,
    markersize=None,
    label="x^*",
    drop_dashes=True,   # dashed lines to axes
    show_ray=False,     # expansion path
    drop_stroke=None,
    ray_stroke=None,
    marker=None,
)
```

Mark the optimal bundle returned by `solve()`.

`drop_dashes`
bool · default `True`
Drop lines to both axes.

`show_ray`
bool · default `False`
Draw the expansion path through the origin and the optimum.

`marker`
Marker | None · default `None`
Style of the equilibrium marker (Section 4.3).

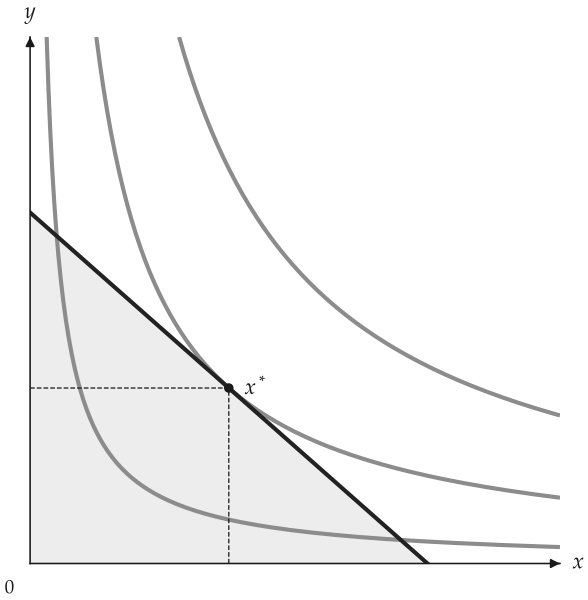


Figure 4
Equilibrium and drop lines.

Example 4

```python
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)  # the optimum and dashed guides
cvs.save("add_equilibrium.png")
```

`add_ray`

```
cvs.add_ray(
    slope,              # dy/dx
    color=None,
    linewidth=None,
    stroke=None,
)
```

Draw a ray from the origin with slope `slope` ($\mathrm{d}y/\mathrm{d}x$), for an expansion path or a fixed consumption ratio.

Example 5

```python
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.add_ray(eq.y / eq.x)  # a ray through the optimum
cvs.save("add_ray.png")
```

`add_point`

```
cvs.add_point(
    x, y,
    label=None,
    color=None,
    markersize=None,
```

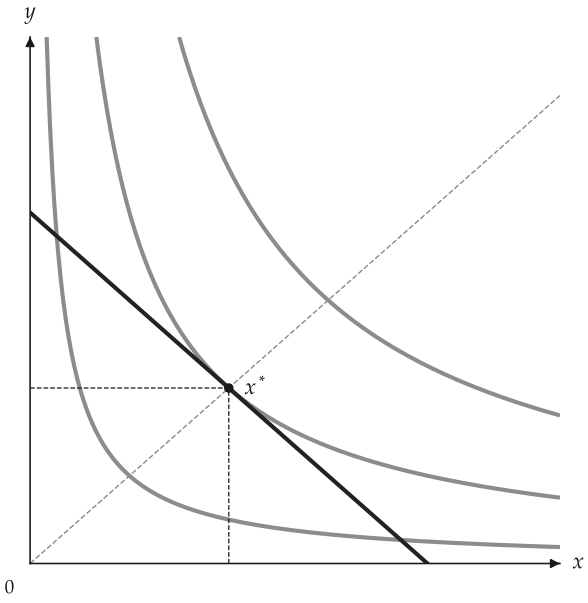


Figure 5
Expansion-path ray.

```
    offset=None,        # label offset in pt
    marker=None,
)
```

Mark any point, such as a bundle to compare with the optimum.

`label` — *str | Label | None* · default `None`
Marker text, as a string or a `Label`.

`offset` — *tuple | None* · default `None`
Offset of the text from the point, in points.

`marker` — *Marker | None* · default `None`
Style of the point marker.

Example 6

```
from utility_viz import Canvas, Config, Label, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
cvs.add_utility(model, levels=[5.0])  # one indifference curve u = 5
cvs.add_budget(2.0, 3.0, 30.0)
# it cuts the budget line twice
# a Label puts text by the point
cvs.add_point(3.17, 7.89, label=Label(text="A", position="top-right"))
cvs.add_point(11.83, 2.11, label=Label(text="B", position="top-right"))
cvs.save("add_point.png")
```

`show` `save`

```
cvs.show()                # interactive window
cvs.save("figure.png")    # .png / .pdf / .svg / .tex
```

`show()` opens an interactive window. `save()` writes the file in the format given by its extension (Section 10) and releases the matplotlib figure, so call it last.

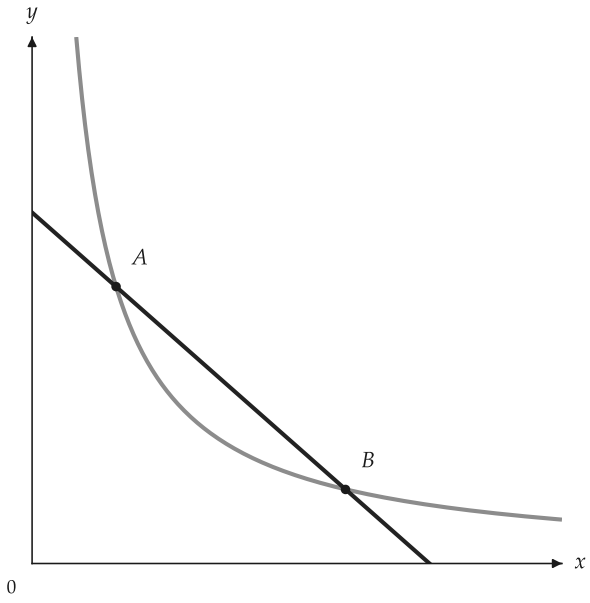


Figure 6
An indifference curve meets the budget line at two points, A and B.

Example 7

```python
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")  # italic x and y
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq, show_ray=True)  # equilibrium with the expansion path
cvs.save("show_save.png")
```

4.3 Styles

`LineStyle` defines line patterns, `ArrowStyle` arrowheads, and `Stroke` bundles width, pattern, colour and arrowhead for axes, budget lines, paths and drop lines.

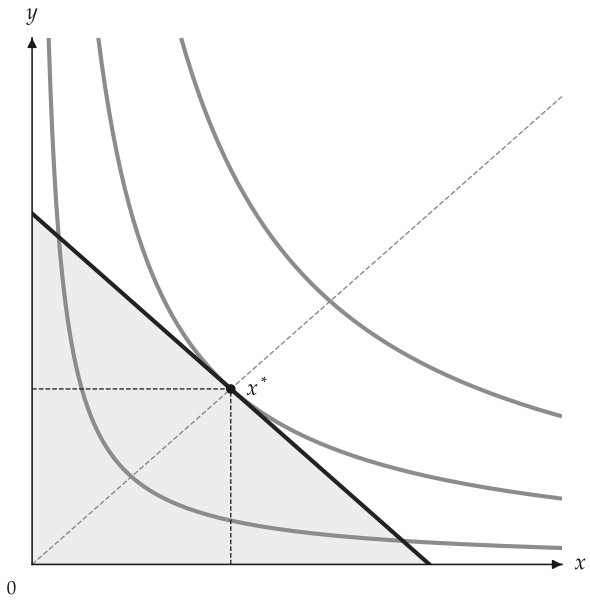


Figure 7
The complete canvas.

Line styles

`LineStyle` `LineStyle.SOLID, DASHED, DOTTED, DASHDOT`

New: v1.7.0 Solid, dashed, dotted and dash-dot lines (Figure 8). Axes accept the enum or the matching string; other lines use `Stroke(style=...)`. `LineStyle` is MosaicKit's `DashStyle` under its old name.

`SOLID` *LineStyle* Solid line.

`DASHED` *LineStyle* Dashed line.

`DOTTED` *LineStyle* Dotted line.

`DASHDOT` *LineStyle* Dash-dot line.

Example 8

```
import matplotlib.pyplot as plt

from utility_viz import Canvas, Config, LineStyle

Config.load("manual.toml").use()  # the settings behind this manual's figures

styles = [LineStyle.SOLID, LineStyle.DASHED, LineStyle.DOTTED, LineStyle.DASHDOT]

fig, axes = plt.subplots(1, 4, figsize=(7.6, 2.0))
for ax, style in zip(axes, styles):
    Canvas(
        title=style.value,
        x_label="$x$",
        y_label="$y$",
        x_line_style=style,
        y_line_style=style,
        fig=fig,
        ax=ax,
    )

# leave room for axis labels
fig.subplots_adjust(wspace=0.62)
fig.savefig("line_styles.png", dpi=160)
```

Arrow styles

`ArrowStyle` `ArrowStyle.SIMPLE, TRIANGLE, FANCY, WEDGE`

New: v1.7.0 Four arrowhead styles (Figure 9). Axes use `x_arrow_style` and `y_arrow_style`; other lines use `Stroke(arrow=...)`. `ArrowStyle` is MosaicKit's; `SIMPLE` is its `OPEN`, and `"->"` and `"-|>"` are still accepted.

`SIMPLE` *ArrowStyle* Open arrowhead (`->`).

`TRIANGLE` *ArrowStyle* Filled triangle (`-|>`), the default.

`FANCY` *ArrowStyle* Curved, tapering arrowhead.

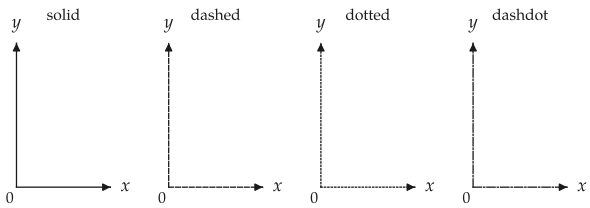


Figure 8
The four line styles.

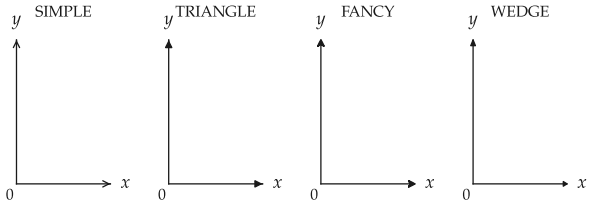


Figure 9
The four arrow styles.

`WEDGE` Thin wedge-shaped arrowhead.
ArrowStyle

Example 9

```
import matplotlib.pyplot as plt

from utility_viz import ArrowStyle, Canvas, Config

Config.load("manual.toml").use()  # the settings behind this manual's figures

styles = [ArrowStyle.SIMPLE, ArrowStyle.TRIANGLE, ArrowStyle.FANCY, ArrowStyle.WEDGE]

fig, axes = plt.subplots(1, 4, figsize=(7.6, 2.0))
for ax, style in zip(axes, styles):
    Canvas(
        title=style.name,
        x_label="$x$",
        y_label="$y$",
        x_arrow_style=style,
        y_arrow_style=style,
        fig=fig,
        ax=ax,
    )

# leave room for axis labels
fig.subplots_adjust(wspace=0.62)
fig.savefig("arrow_styles.png", dpi=160)
```

Stroke

`Stroke` `Stroke(width=⟨float⟩, style=⟨LineStyle⟩, color=⟨str⟩, arrow=⟨ArrowStyle⟩, opacity=⟨float⟩)`

New: v1.7.0 Width, pattern, colour, arrowhead and opacity of one line. Fields left unset inherit from
Updated: v1.9.0 the active theme. `opacity` ranges from 0 to 1.

Axis

`Axis` `Axis(label=None, label_position=None, stroke=None)`

New: v1.8.0 Label, label position and line of one axis in one object. `Canvas`, `Figure`, `DemandDiagram` and `EdgeworthBox` all accept `x_axis` and `y_axis`. When shorthand arguments are passed as well, fields set in the `Axis` win.

```
from utility_viz import Axis, Canvas, Label, Stroke

cvs = Canvas(
    x_axis=Axis(
        label=Label(text="x_1", fontsize=16),
        label_position="bottom",
        stroke=Stroke(width=1.2),
    ),
    y_axis=Axis(label="x_2"),
)
```

Marker and Label

`Marker` `Marker(color=None, size=None, shape=None, opacity=None)`

New: v1.8.0 Colour, size, shape and opacity of a point marker. `shape` takes a matplotlib marker string
Updated: v1.9.0 such as `"o"`, `"s"`, `"^"`, `"D"` or `"*"`.

`Label` `Label(text=None, position=None, offset=None, color=None, fontsize=None, visible=None, opacity=None)`

New: v1.8.0 *Updated: v1.9.0* Text, position, offset, colour, size, visibility and opacity of a label. The position is a side or a corner; `Label(visible=False)` hides the text.

```
from utility_viz import Label, Marker

cvs.add_equilibrium(
    eq,
    marker=Marker(shape="s", size=8),
    label=Label(position="bottom-left", offset=8),
)
cvs.add_point(
    12, 2,
    marker=Marker(color="black", shape="D"),
    label=Label(text="A", position="left", fontsize=14),
)
```

Fill

`Fill` `Fill(color=None, opacity=None)`

New: v1.8.0 *Updated: v1.9.0* Colour and opacity of a shaded area. `alpha` is a compatible shorthand for `opacity`; one call must not pass different values for both.

```
from utility_viz import Fill

cvs.add_budget(
    2, 3, 30,
    fill=Fill(color="lightgrey", opacity=0.4),
)
```

4.4 Method chaining

The `add_*` methods can be chained, ending with `save()`:

```
Canvas(x_max=20, y_max=15) \
    .add_utility(model, levels=lvls) \
    .add_budget(2.0, 3.0, 30.0, fill=True) \
    .add_equilibrium(eq, show_ray=True) \
    .save("figure.png")
```

5 Figures and demand diagrams

On top of `Canvas`, utility-viz builds these larger diagrams:

- `Figure` for multi-panel layouts;
- `PricePath` and `IncomePath` for budget and equilibrium sweeps;
- `DemandDiagram` for linked goods-space and Marshallian-demand views;
- `decompose_price_effect` for the Hicks and Slutsky decompositions;
- `EdgeworthBox` for two-consumer exchange diagrams.

5.1 Multi-panel figures

`Figure` `Figure(⟨layout⟩, x_max=⟨float⟩, y_max=⟨float⟩, ..., shared_x=False, shared_y=False)`

New: v1.2.0 Compose several canvases into one figure, for before-and-after comparisons, decompositions or classroom slides. It takes the same styling arguments as `Canvas`. `fig[idx]` returns the panel `Canvas`, so the drawing API is unchanged once the layout exists.

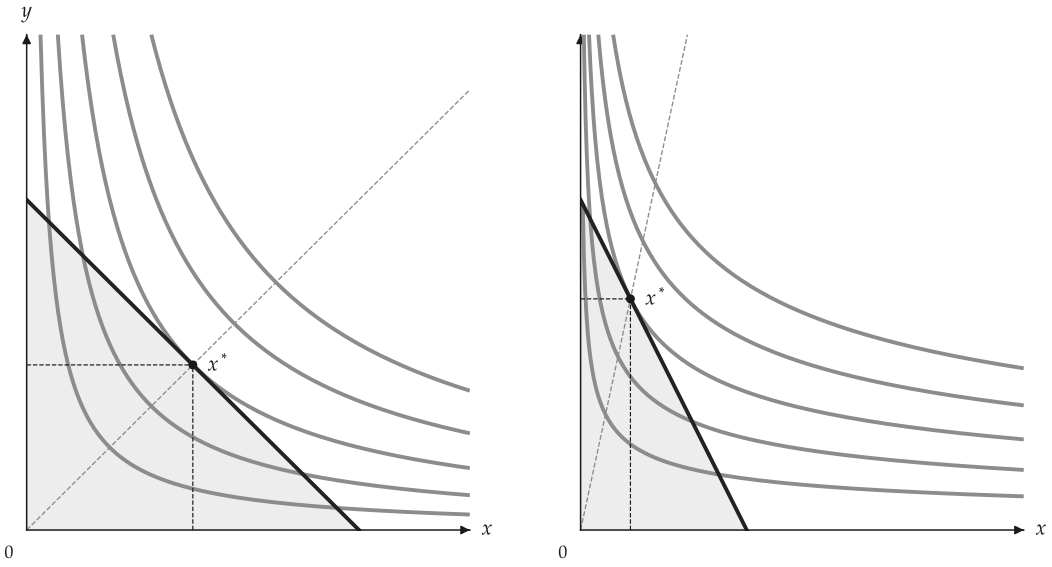


Figure 10
Price change, before and after.

Example 10

```python
from utility_viz import Config, Figure, Layout, levels, solve
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

fig = Figure(Layout.SIDE_BY_SIDE, x_max=20, y_max=15, x_label="$x$", y_label="$y$",
shared_y=True)

# before and after x doubles
cases = [
    (CobbDouglas(alpha=0.5, beta=0.5), 2.0, 3.0, 30.0),
    (CobbDouglas(alpha=0.3, beta=0.7), 4.0, 3.0, 30.0),
]
for idx, (model, px, py, income) in enumerate(cases):
    eq = solve(model, px=px, py=py, income=income)
    panel = fig[idx]  # each panel is a Canvas
    panel.add_utility(model, levels=levels.around(eq.utility, n=5))
    panel.add_budget(px, py, income, fill=True)
    panel.add_equilibrium(eq, show_ray=True)
fig.save("side_by_side.png")
```

`Layout`
New: v1.2.0
The available layouts: `SINGLE`, `STACKED`, `SIDE_BY_SIDE`, `TOP_TWO_BOTTOM_ONE`, `TOP_ONE_BOTTOM_TWO`, `GRID_2X2` and `GRID_3X3`.

`SINGLE`
Layout
One panel.

`STACKED`
Layout
Two panels, one above the other.

`SIDE_BY_SIDE`
Layout
Two panels, side by side.

`TOP_TWO_BOTTOM_ONE`
Layout
Two panels on top, one below.

`TOP_ONE_BOTTOM_TWO`
Layout
One panel on top, two below.

`GRID_2X2`
Layout
Four panels in a 2 × 2 grid.

`GRID_3X3`
Layout
Nine panels in a 3 × 3 grid.

5.2 Paths

PricePath
IncomePath

```
from utility_viz import IncomePath, LinearBudget, PricePath
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
budget = LinearBudget(px=2.0, py=2.0, income=40.0)

price_path = PricePath(
    model,
    budget=budget,
    price="px",
    price_range=(0.8, 6.0),
    n=40,
)
income_path = IncomePath(
    model,
    budget=budget,
    income_range=(20.0, 80.0),
    n=30,
)
```

New: v1.2.0 Sweep one budget parameter while repeatedly solving the consumer's problem. Use a path to draw price- and income-consumption curves with `Canvas.add_path`, to feed a `Demand Diagram`, or to inspect how bundles move as prices or income vary.

Canvas.add_path

```
from utility_viz import Canvas, levels

eq = price_path.equilibria[len(price_path.equilibria) // 2]
lvls = levels.around(eq.utility, n=5)

Canvas(x_max=25, y_max=20) \
    .add_utility(model, levels=lvls) \
    .add_path(price_path, label="PCC") \
    .save("price_path.png")
```

New: v1.2.0 Draw a path directly on a canvas when a full demand diagram is not needed. For Cobb-Douglas preferences the price-consumption curve is horizontal (Figure 11): spending on y does not depend on p_x.

Example 11

```
from utility_viz import Canvas, Config, LinearBudget, PricePath, levels
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
budget = LinearBudget(px=2.0, py=2.0, income=40.0)
path = PricePath(model, budget=budget, price="px", price_range=(0.8, 6.0), n=40)

# curves around the middle optimum
eq = path.equilibria[len(path.equilibria) // 2]

cvs = Canvas(x_max=25, y_max=20, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_path(path)  # the price-consumption curve
cvs.save("price_path.png")
```

5.3 Demand diagrams

DemandDiagram

`DemandDiagram(⟨price path⟩, title=None, ...)`
`.add_marshallian_panel(price_markers=None, show_pcc=False, show_demand_guides=True)`

New: v1.2.0 A stacked two-panel figure: indifference curves, budget lines and equilibria in the top panel, the corresponding Marshallian demand curve in the bottom one (Figure 12).

- `DemandDiagram` expects a `PricePath`.

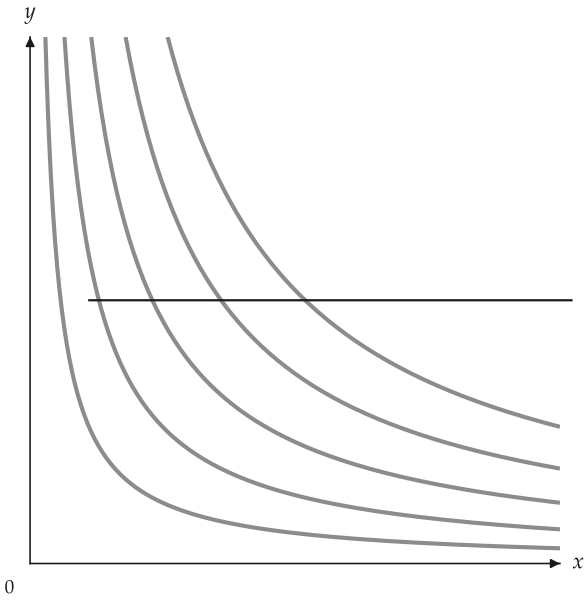


Figure 11
The price-consumption curve as p_x varies.

- Smooth, kinked and corner demand are handled differently, so that the bottom panel stays economically meaningful.
- `show_pcc=True` overlays the price-consumption curve in goods space.

Example 12

```
from utility_viz import DemandDiagram, LinearBudget, PricePath
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
budget = LinearBudget(px=2.0, py=2.0, income=40.0)
path = PricePath(
    model,
    budget=budget,
    price="px",
    price_range=(0.8, 6.0),
    n=40,
)

fig = DemandDiagram(path, title="Demand: Cobb-Douglas")
fig.add_marshallian_panel(
    price_markers=[1.5, 4.0],
    show_pcc=False,
    show_demand_guides=True,
)
fig.save("demand_cobb_douglas.png")
```

The figure below is drawn from MosaicKit layers, so each label can be placed by hand.

Example 13

```
from mosaickit import (
    AxisSpec,
    Canvas,
    CanvasGrid,
    CanvasSpec,
    DashStyle,
    Fill,
    Marker,
    MarkerLayer,
    PathLayer,
    Stroke,
    StyleBundle,
    TextLayer,
    TextStyle,
    build_axes,
)
import numpy as np

from utility_viz import solve
from utility_viz.core.scenes import UTILITY_THEME, budget_layers, indifference_layers
from utility_viz.models import CobbDouglas

INK, DARK, GREY = "#1a1a1a", "#222222", "#8c8c8c"  # the colours of manual.toml
text = TextStyle(color=INK, size=14)

# Cobb-Douglas demand for x
model = CobbDouglas(alpha=0.5, beta=0.5)
py, income, px_a, px_b = 2.0, 40.0, 1.5, 4.0
points = {"A": px_a, "B": px_b}
optimum = {name: solve(model, px=px, py=py, income=income) for name, px in points.items()}
X = 29.0  # x runs from 0 to 29


def panel(y_min, y_max, height, y_label):
    """A white panel with thin axes; guides from a point to the axes are dashed."""
    spec = CanvasSpec(
        x_range=(0, X), y_range=(y_min, y_max), width=4.6, height=height, x_label="",
y_label=""
    )
    roles = {
        "canvas": StyleBundle(fill=Fill(color="#ffffff", opacity=1.0)),
        "axes": StyleBundle(stroke=Stroke(color=INK, width=1.0), text=text),
        "guide": StyleBundle(stroke=Stroke(color=INK, width=0.8, dash=DashStyle.DASHED)),
    }
    cv = Canvas(spec, theme=UTILITY_THEME, role_overrides=roles)
    cv.extend(
        build_axes(
            AxisSpec((0, X), arrow="end", label="$x$"),
            AxisSpec((0, y_max), arrow="end", label=y_label),
        )
    )
    cv.add(TextLayer((0, 0), "0", offset=(-6, -8), anchor="top-right", style=text,
id="origin"))
    return cv


def mark(cv, name, x, y):
    """A black dot with its italic letter, and dashed guides to both axes."""
    cv.add(PathLayer([(0, y), (x, y)], id=f"{name}.h", role="guide"))
    cv.add(PathLayer([(x, 0), (x, y)], id=f"{name}.v", role="guide"))
    cv.add(
        MarkerLayer(
            [(x, y)], marker=Marker(color=INK, size=26, edge_width=0), id=f"{name}.dot",
z_index=5
        )
    )
    cv.add(
        TextLayer(
            (x, y),
            name,
            math=True,
            offset=(8, 6),
            anchor="bottom-left",
            style=text,
            id=f"{name}.label",
        )
    )


# top panel is the goods space
top = panel(-3.5, 24, 3.9, "$y$")
top.extend(budget_layers(px_a, py, income, layer_id="ba", stroke=Stroke(color=DARK,
width=2.6)))
top.extend(
    budget_layers(
        px_b, py, income, layer_id="bb", stroke=Stroke(color=DARK, width=2.2,
dash=DashStyle.DASHED)
    )
)
for name, eq in optimum.items():
    curve, _ = indifference_layers(
```

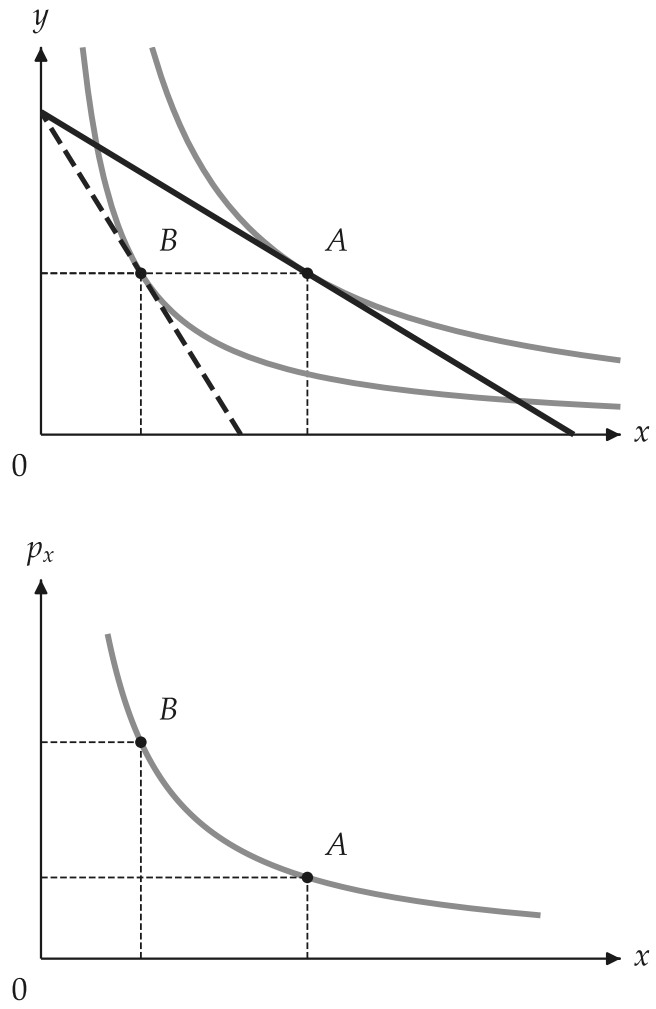


Figure 12
Demand curve at the optima.

5.4 Price-effect decomposition

`decompose_price_effect` `decompose_price_effect(`⟨*model*⟩`, px=(`⟨*old*⟩`, `⟨*new*⟩`), py=`⟨*float*⟩`, income=`⟨*float*⟩`, method=`⟨*method*⟩`)`

New: v1.5.0 Separate a price change into substitution and income effects. Choose `Decomposition Method.HICKS` to hold the original utility fixed (Hicks, 1939), or `SLUTSKY` (Slutsky, 1915) to keep the original bundle affordable. The result has these fields:

`A, B, C` The original, compensated and final bundles.

`substitution_effect` Substitution effect.

`income_effect` Income effect.

`total_effect` Total effect.

`compensated_income` Income after compensation.

`DecompositionMethod` `DecompositionMethod.HICKS, SLUTSKY`

New: v1.5.0 How the compensated bundle B is chosen.

`HICKS` *DecompositionMethod* Hold the original utility fixed: B lies on the original indifference curve.

`SLUTSKY` *DecompositionMethod* Keep the original bundle affordable: B is the optimum on the budget line through A at the new prices.

`Canvas.add_decomposition`
New: v1.5.0
Updated: v1.9.0

Draw the three bundles A, B and C, the budget lines, the indifference curves and the effect arrows. A Hicks decomposition draws the curves through A and C; a Slutsky decomposition also draws the one through B.

`show_curves`
bool · default `True`

Draw the decomposition's indifference curves; set to `False` when you draw your own.

`curve_stroke`
Stroke | *None* · default `None`

Style of the indifference curves.

`curve_label`
Label | *None* · default `None`

Show and style the U_0, U_1 and U_B labels.

`substitution, income`
Effect | *None* · default `None`

Style of the substitution and income effects (see `Effect` below).

`point_marker`
Marker | *None* · default `None`

Markers of the bundles.

`point_label`
Label | *None* · default `None`

Text of the bundles.

`legend`
Legend | *None* · default `None`

Position and style of the legend (see `Legend` below).

`Effect`

`Effect(color=None, y=None, label=None, label_position="right", label_offset=4, opacity=None)`

New: v1.8.0
Updated: v1.9.0

Colour, height of the arrow below the horizontal axis, label and opacity of the substitution or income effect. The line itself can still be overridden with `substitution_stroke` or `income_stroke`.

`Legend`

`Legend(position="auto", fontsize=None, frame=None, columns=None, visible=None, opacity=None)`

New: v1.9.0

`"auto"` picks the inner corner that hides the least of the plot; when all four corners overlap it, the legend moves to the right. Inner corners such as `"upper left"` and outer positions `"top"`, `"bottom"`, `"left"` and `"right"` can also be given. `Legend(visible=False)` hides the legend.

Example 14

```
from utility_viz import (
    Canvas,
    DecompositionMethod,
    Effect,
    Label,
    Legend,
    decompose_price_effect,
)
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
result = decompose_price_effect(
    model,
    px=(2.0, 4.0),
    py=3.0,
    income=60.0,
    method=DecompositionMethod.HICKS,
)

(
    Canvas(x_max=25, y_max=25, title="Hicks decomposition")
    .add_decomposition(
        result,
        show_arrows=True,
        label_effects=True,
        show_x_projections=True,
        substitution=Effect(label="SE"),
        income=Effect(label="IE", opacity=0.8),
        curve_label=Label(position="right"),
        legend=Legend(position="bottom"),
    )
    .save("hicks.png")
)
```

The figure below is also drawn from MosaicKit layers; the two effects are named under the axis.

Example 15

```
from mosaickit import (
    ArrowLayer,
    ArrowStyle,
    AxisSpec,
    Canvas,
    CanvasSpec,
    DashStyle,
    Fill,
    Marker,
    MarkerLayer,
    PathLayer,
    Stroke,
    StyleBundle,
    TextLayer,
    TextStyle,
    build_axes,
)

from utility_viz import DecompositionMethod
from utility_viz.core.scenes import UTILITY_THEME, budget_layers, indifference_layers
from utility_viz.models import CobbDouglas, decompose_price_effect

INK, DARK, GREY = "#1a1a1a", "#222222", "#8c8c8c"  # the colours of manual.toml
text = TextStyle(color=INK, size=14)
effect_name = TextStyle(color=INK, size=13)

# p_x doubles from 2 to 4
# A old and C new and B compensated
model = CobbDouglas(alpha=0.5, beta=0.5)
px0, px1, py, income = 2.0, 4.0, 3.0, 60.0
result = decompose_price_effect(
    model, px=(px0, px1), py=py, income=income, method=DecompositionMethod.HICKS
)
bundles = {"A": result.A, "B": result.B, "C": result.C}
X, Y = 26.0, 24.0

# a white canvas with thin axes
spec = CanvasSpec(x_range=(0, X), y_range=(-7.5, Y), width=4.6, height=4.2, x_label="",
y_label="")
roles = {
    "canvas": StyleBundle(fill=Fill(color="#ffffff", opacity=1.0)),
    "axes": StyleBundle(stroke=Stroke(color=INK, width=1.0), text=text),
    "guide": StyleBundle(stroke=Stroke(color=INK, width=0.8, dash=DashStyle.DASHED)),
}
cv = Canvas(spec, theme=UTILITY_THEME, role_overrides=roles)
cv.extend(
    build_axes(
        AxisSpec((0, X), arrow="end", label="$x$"), AxisSpec((0, Y), arrow="end",
label="$y$")
    )
)
cv.add(TextLayer((0, 0), "0", offset=(-6, -8), anchor="top-right", style=text, id="origin"))

# three budget lines and two curves
cv.extend(budget_layers(px0, py, income, layer_id="b0", stroke=Stroke(color=DARK,
width=2.6)))
cv.extend(budget_layers(px1, py, income, layer_id="b1", stroke=Stroke(color=DARK,
width=2.6)))
compensated = [(0, result.compensated_income / py), (X, (result.compensated_income - px1 *
X) / py)]
cv.add(
    PathLayer(
        compensated, stroke=Stroke(color=INK, width=1.0, dash=DashStyle.DASHED),
id="compensated"
    )
)
for name in ("A", "C"):
    curve, _ = indifference_layers(
        model,
        [bundles[name].utility],
        X,
        Y,
        layer_id=f"u.{name}",
        stroke=Stroke(color=GREY, width=2.6),
    )
    cv.extend(curve)

# three bundles with dashed guides
for name, bundle in bundles.items():
    cv.add(PathLayer([(bundle.x, 0), (bundle.x, bundle.y)], id=f"{name}.guide",
role="guide"))
    cv.add(
        MarkerLayer(
            [(bundle.x, bundle.y)],
            marker=Marker(color=INK, size=26, edge_width=0),
            id=f"{name}.dot",
            z_index=5,
        )
    )
    cv.add(
        TextLayer(
```

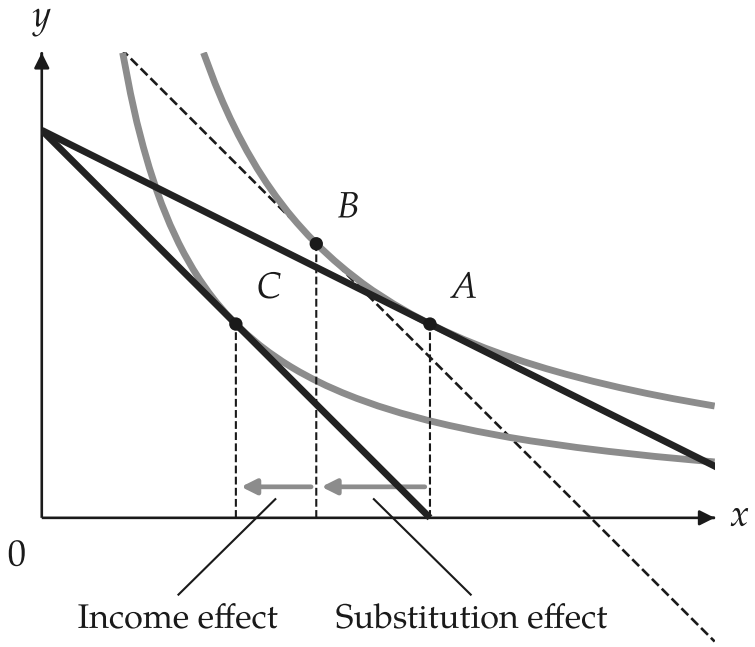


Figure 13
Hicks decomposition.

5.5 Edgeworth box

`EdgeworthBox` `EdgeworthBox(⟨utility A⟩, ⟨utility B⟩, total_x=⟨float⟩, total_y=⟨float⟩, ...)`

New: v1.3.0

A two-consumer exchange economy (Edgeworth, 1881). Consumer A is measured from the lower-left origin and consumer B from the upper-right origin. Layers are added with `add_endowment`, `add_contract_curve`, `add_core`, `add_price_line` and `add_walrasian_equilibrium`.

Use `method="mrs"` for the contract curve of smooth preferences, and `method="pareto"` for models with kinks, corners or custom piecewise utility functions.

Example 16

```
from utility_viz import EdgeworthBox
from utility_viz.models import CobbDouglas

box = EdgeworthBox(
    CobbDouglas(alpha=0.8, beta=0.2),
    CobbDouglas(alpha=0.2, beta=0.8),
    total_x=12.0,
    total_y=10.0,
    title="Asymmetric Cobb-Douglas",
)

(
    box.add_endowment(5.0, 4.0)
    .add_contract_curve(n=100, method="mrs")
    .add_core()
    .add_price_line(px=1.2, py=1.0)
    .add_walrasian_equilibrium(px=1.2, py=1.0)
    .show_legend(loc="center left", bbox_to_anchor=(1.02, 0.5))
    .save("edgeworth.png")
)
```

The figure below is drawn from MosaicKit layers after a textbook figure, with both consumers' indifference curves.

Example 17

```
import numpy as np
from mosaickit import (
    ArrowLayer,
    ArrowStyle,
    AxisSpec,
    Canvas,
    CanvasSpec,
    Fill,
    FillLayer,
    Marker,
    MarkerLayer,
    PathLayer,
    Stroke,
    StyleBundle,
    TextLayer,
    TextStyle,
)
from scipy.interpolate import CubicSpline

from utility_viz.core.scenes import UTILITY_THEME

INK, GREY, LENS = "#1a1a1a", "#8c8c8c", "#ececec"  # the colours of manual.toml
text = TextStyle(color=INK, size=12)

# after a textbook figure
# J from O_J and K from O_K
# curves are splines through control points
J = {  # J's control points low to high
    1: (
        [3.80, 3.91, 4.18, 4.59, 5.12, 5.75, 6.46, 7.16, 8.07],
        [5.15, 4.29, 3.49, 2.74, 2.08, 1.52, 1.07, 0.79, 0.58],
    ),
    2: (
        [4.26, 4.37, 4.60, 4.94, 5.25, 5.90, 6.30, 7.15, 7.84],
        [5.28, 4.55, 3.86, 3.22, 2.79, 2.13, 1.83, 1.38, 1.17],
    ),
    3: (
        [5.12, 5.20, 5.38, 5.64, 5.95, 6.35, 6.81, 7.29, 7.80],
        [5.40, 4.88, 4.39, 3.93, 3.50, 3.18, 2.91, 2.73, 2.63],
    ),
}
K = {  # K's from outer to inner
    1: (
        [3.38, 4.17, 4.91, 5.57, 5.95, 6.60, 6.93, 7.16, 7.19],
        [4.86, 4.65, 4.30, 3.83, 3.50, 2.57, 1.83, 0.79, 0.24],
    ),
    2: (
        [3.62, 4.22, 4.77, 5.27, 5.68, 6.03, 6.30, 6.47, 6.54],
        [4.35, 4.14, 3.83, 3.43, 2.97, 2.44, 1.83, 1.26, 0.64],
    ),
    3: (
        [3.64, 4.12, 4.55, 4.93, 5.25, 5.53, 5.73, 5.86, 5.92],
        [3.80, 3.65, 3.41, 3.11, 2.79, 2.33, 1.88, 1.41, 0.92],
    ),
}
# four allocations A to D
allocations = {"A": (7.16, 0.79), "B": (6.30, 1.83), "C": (5.25, 2.79), "D": (5.95, 3.50)}
# positions of the labels
letters = {"A": (7.41, 0.93), "B": (6.55, 1.90), "C": (5.48, 2.93), "D": (6.14, 3.67)}
curve_labels = {
    "U_J^1": (8.35, 0.48),
    "U_J^2": (8.12, 1.09),
    "U_J^3": (8.05, 2.58),
    "U_K^1": (7.46, 0.25),
    "U_K^2": (6.58, 0.36),
    "U_K^3": (5.95, 0.66),
}


def spline(points, n=240):
    """A smooth curve through the control points, as n sampled points."""
    points = np.column_stack(points)  # turn xs and ys into points
    chord = np.r_[0, np.cumsum(np.hypot(*np.diff(points, axis=0).T))]
    return CubicSpline(chord, points)(np.linspace(0, chord[-1], n))


curves_j = {k: spline(v) for k, v in J.items()}
curves_k = {k: spline(v) for k, v in K.items()}

# box plus margin for the arrows
# 0.6 inch per unit
x_range, y_range, unit = (-1.0, 11.15), (-0.8, 6.8), 0.6
spec = CanvasSpec(
    x_range=x_range,
    y_range=y_range,
    x_label="",
    y_label="",
    width=(x_range[1] - x_range[0]) * unit / 0.775,
    height=(y_range[1] - y_range[0]) * unit / 0.77,
)
roles = {"canvas": StyleBundle(fill=Fill(color="#ffffff", opacity=1.0))}
```

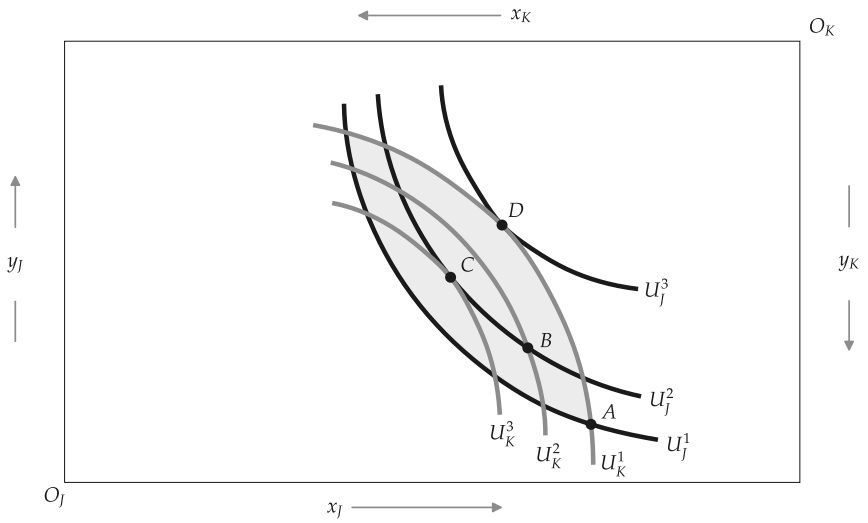


Figure 14
Both consumers' indifference curves through allocations A to D; the shaded lens holds the allocations both prefer to A.

6 Core models

The core models cover smooth, kinked, linear, income-dependent, and reference-point preferences. They all live in `utility_viz.models`. This chapter gives each model's function, parameters, distinct behaviour, and figure.

6.1 Smooth preferences

These models produce smooth indifference curves and normally have an interior optimum when prices and income are positive.

Cobb-Douglas

`CobbDouglas` `CobbDouglas(alpha=0.5, beta=0.5)`

The standard model for smooth, strictly convex preferences (Cobb & Douglas, 1928):

$$u(x, y) = x^{\alpha} y^{\beta}.$$

The exponents control the relative weight placed on each good. The indifference curves approach both axes without touching them.

`alpha` Weight on good x.
default `0.5`

`beta` Weight on good y.
default `0.5`

CES

`CES` `CES(alpha=0.5, beta=0.5, rho=0.5)`

Constant elasticity of substitution (Arrow et al., 1961): the ease of substitution varies while preferences stay smooth,

$$u(x, y) = (\alpha x^{\rho} + \beta y^{\rho})^{1/\rho}.$$

The elasticity of substitution is $\sigma = 1/(1-\rho)$. As ρ changes, CES approaches Cobb-Douglas ($\rho \to 0$), Leontief ($\rho \to -\infty$) or perfect substitutes ($\rho \to 1$).

`alpha` Weight on good x.
default `0.5`

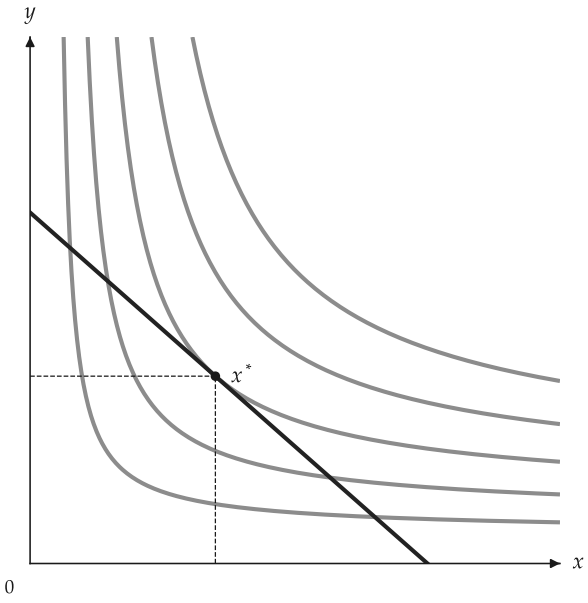


Figure 15
CES with $\rho = -0.5$.

`beta` Weight on good y.
default `0.5`

`rho` Substitution parameter, $\rho \neq 1$.
default `0.5`

Example 18

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CES

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CES(alpha=0.5, beta=0.5, rho=-0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("ces.png")
```

Translog

`Translog` `Translog(alpha_0=0.0, alpha_x=0.5, alpha_y=0.5, beta_xx=0.0, beta_yy=0.0, beta_xy=0.0)`

New: v1.1.0 A flexible log-quadratic model for smooth preferences (Christensen et al., 1975):

$$\ln u(x,y) = \alpha_0 + \alpha_x \ln x + \alpha_y \ln y + \frac{1}{2}\beta_{xx}(\ln x)^2 + \frac{1}{2}\beta_{yy}(\ln y)^2 + \beta_{xy} \ln x \ln y.$$

The quadratic and interaction terms let the curvature vary across the consumption space. With all β coefficients zero it reduces to a Cobb-Douglas-style log-linear form.

`alpha_0` Log-utility intercept.
default `0.0`

`alpha_x` First-order weight on $\ln x$.
default `0.5`

`alpha_y` First-order weight on $\ln y$.
default `0.5`

`beta_xx` Curvature in x.
default `0.0`

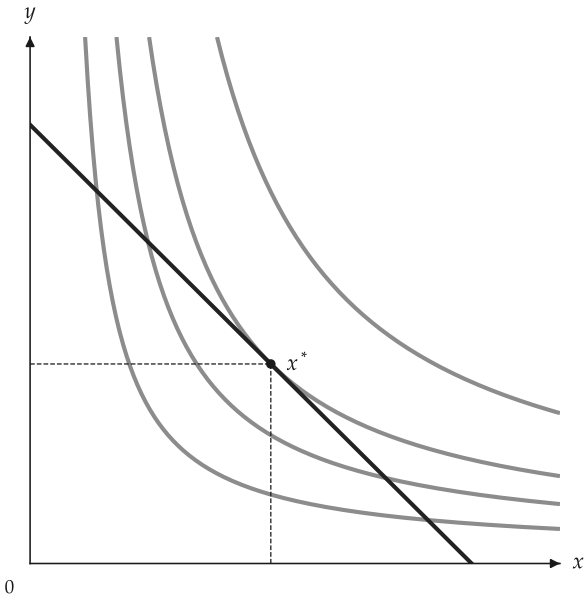


Figure 16
Translog with $\beta_{xy} = 0.12$.

`beta_yy` Curvature in y.
default `0.0`

`beta_xy` Interaction between x and y.
default `0.0`

Example 19

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import Translog

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = Translog(alpha_x=0.6, alpha_y=0.4, beta_xy=0.12)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=18, y_max=12, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=4))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("translog.png")
```

6.2 Kinks and corners

These models show how non-smooth or linear preferences move the optimum to a kink or a corner.

Perfect complements

`Leontief` `Leontief(a=1.0, b=1.0)`

Goods consumed in fixed proportions:

$$u(x, y) = \min(ax, by).$$

The indifference curves are L-shaped, and the optimum lies at the kink where the two weighted quantities are equal.

The corners are drawn as exact right angles, in image and TikZ output alike; this also holds for piecewise `min`/`max` functions built with `CustomUtility`.

`a` Weight on good x.
default `1.0`

`b` Weight on good y.
default `1.0`

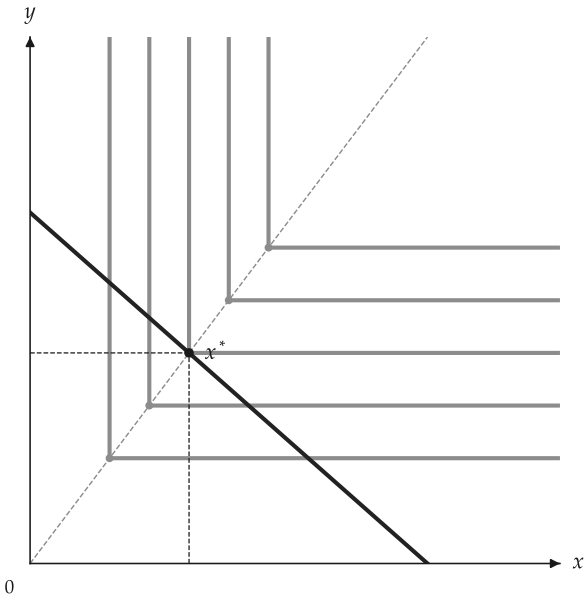


Figure 17
Leontief, with the ray through the kinks.

Example 20

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import Leontief

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = Leontief(a=1.0, b=1.0)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(
    model,
    levels=levels.around(eq.utility, n=5),
    show_rays=True,
    show_kinks=True,  # ray and marker at each kink
)
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("leontief.png")
```

Perfect substitutes

`PerfectSubstitutes` `PerfectSubstitutes(a=1.0, b=1.0)`

A constant rate of trade-off between the goods:

$$u(x, y) = ax + by.$$

The indifference curves are straight lines. The consumer normally buys only the good with the greater marginal utility per dollar, a corner solution.

`a` Marginal utility of good x.
default `1.0`

`b` Marginal utility of good y.
default `1.0`

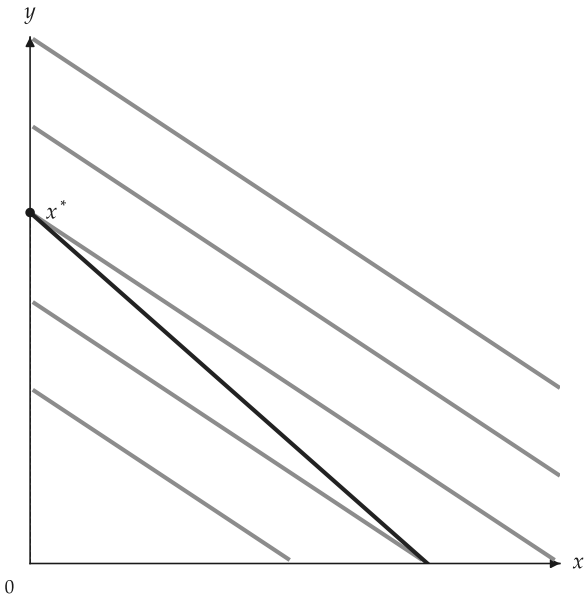


Figure 18
Perfect substitutes: a corner solution on the y-axis.

Example 21

```python
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import PerfectSubstitutes

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = PerfectSubstitutes(a=1.0, b=2.0)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("perfect_substitutes.png")
```

Maximum utility

`maximum` `CustomUtility(func=lambda x, y: np.maximum(a * x, b * y))`

Maximum utility keeps only the larger weighted quantity in each bundle:

$$u(x, y) = \max(ax, by).$$

Each indifference curve has two arms that extend towards the axes. The preferences are non-convex, so on a linear budget the optimum normally lies at the intercept with the greater weighted utility. There is no dedicated class; the model is built with `CustomUtility` (Section 7).

`a` Weight on good x.
default `1.0`

`b` Weight on good y.
default `1.0`

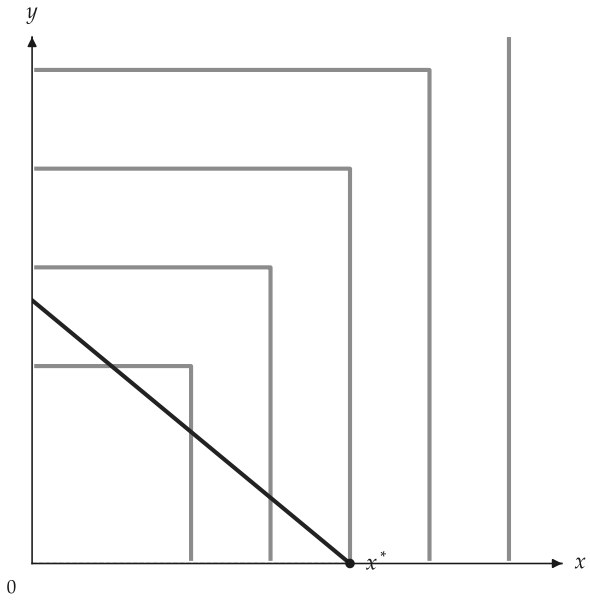


Figure 19
Maximum utility.

Example 22

```python
import numpy as np

from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CustomUtility

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CustomUtility(func=lambda x, y: np.maximum(x, y), name="maximum")
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=25, y_max=20, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("maximum.png")
```

6.3 Income and reference points

These models add special income effects, subsistence requirements or a preferred consumption point.

Quasi-linear

`QuasiLinear` `QuasiLinear(v_func=numpy.log, linear_in="y")`

One good enters utility linearly:

$$u(x, y) = f(x) + y.$$

Once the solution is interior, the non-linear good has no income effect. `linear_in` reverses the roles of x and y.

`v_func` (default `numpy.log`) Increasing, concave function f.

`linear_in` (default `"y"`) Good that enters linearly.

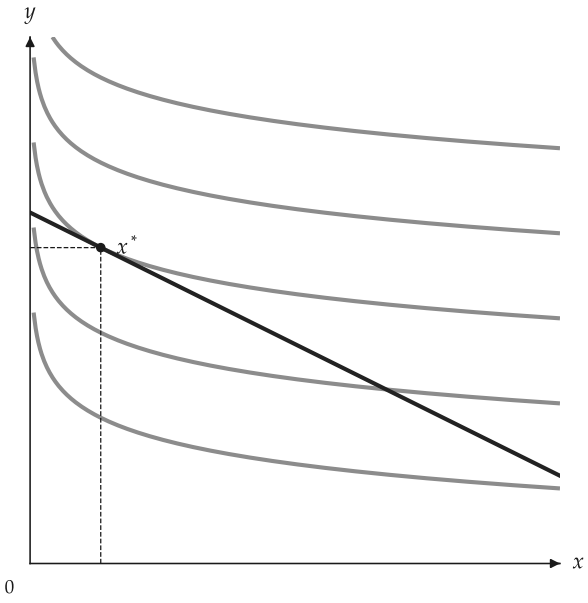


Figure 20
Quasi-linear preferences: vertically parallel indifference curves.

Example 23

```python
import numpy as np

from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import QuasiLinear

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = QuasiLinear(v_func=np.log, linear_in="y")
eq = solve(model, px=1.0, py=2.0, income=20.0)

cvs = Canvas(x_max=15, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(1.0, 2.0, 20.0)
cvs.add_equilibrium(eq)
cvs.save("quasi_linear.png")
```

Haagsma

`Haagsma` `Haagsma(alpha_x=1.0, alpha_y=2.0, gamma_x=2.0, gamma_y=27.0)`

New: v1.9.0

The Haagsma (2012) utility function,

$$u(x,y) = \alpha_x \ln(x - \gamma_x) - \alpha_y \ln\big(\gamma_y - y\big).$$

It is defined for $x > \gamma_x$ and $0 \le y < \gamma_y$, and requires $0 < \alpha_x < \alpha_y$. Marshallian demand for x falls with income, and x is a Giffen good when $\gamma_y p_y < I < \gamma_y p_y + \gamma_x p_x$.

Once income reaches $\gamma_y p_y + \gamma_x p_x$, utility has no finite maximum on the budget line and the model raises `InvalidParameterError`.

`alpha_x` default `1.0` — Weight on good x; must be less than `alpha_y`.

`alpha_y` default `2.0` — Weight on good y.

`gamma_x` default `2.0` — Lower bound on the quantity of x.

`gamma_y` default `27.0` — Upper bound on the quantity of y.

`demand(px, py, income)` — Closed-form interior solution.

`is_giffen(px, py, income)` — Whether x is a Giffen good at the given budget.

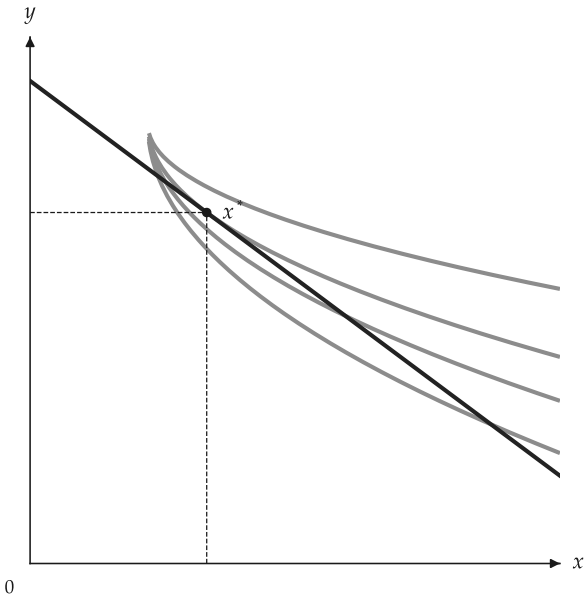


Figure 21
Haagsma with $\gamma_y = 10$.

Example 24

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import Haagsma

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = Haagsma(gamma_y=10.0)
eq = solve(model, px=1.0, py=1.0, income=11.0)

cvs = Canvas(x_max=9, y_max=12, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=4))
cvs.add_budget(1.0, 1.0, 11.0)
cvs.add_equilibrium(eq)
cvs.save("haagsma.png")
```

Stone-Geary

`StoneGeary` `StoneGeary(alpha=0.5, beta=0.5, bar_x=1.0, bar_y=1.0)`

Cobb-Douglas with minimum consumption requirements (Geary, 1950; Stone, 1954):

$$u(x, y) = (x - \bar{x})^{\alpha}(y - \bar{y})^{\beta}.$$

The consumer first covers the subsistence quantities $\bar{x}$ and $\bar{y}$, then allocates the remaining income like a Cobb-Douglas consumer.

`alpha` default `0.5` — Weight on supernumerary x.

`beta` default `0.5` — Weight on supernumerary y.

`bar_x` default `1.0` — Subsistence quantity $\bar{x}$.

`bar_y` default `1.0` — Subsistence quantity $\bar{y}$.

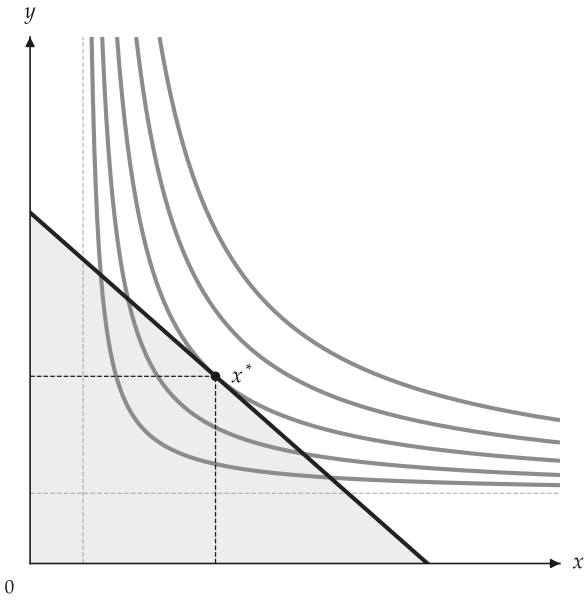


Figure 22
Stone-Geary with $\bar{x} = \bar{y} = 2$.

Example 25

```python
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import StoneGeary

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = StoneGeary(alpha=0.5, beta=0.5, bar_x=2.0, bar_y=2.0)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("stone_geary.png")
```

Satiation

`Satiation` `Satiation(bliss_x=5.0, bliss_y=5.0, a=1.0, b=1.0)`

A bliss point that maximises utility:

$$u(x, y) = -a(x - x^*)^2 - b(y - y^*)^2.$$

Utility falls in every direction away from (x^*, y^*), so the indifference curves are closed ellipses and monotonicity does not hold.

`bliss_x` default `5.0` — Bliss-point coordinate x^*.

`bliss_y` default `5.0` — Bliss-point coordinate y^*.

`a` default `1.0` — Curvature along the x-axis.

`b` default `1.0` — Curvature along the y-axis.

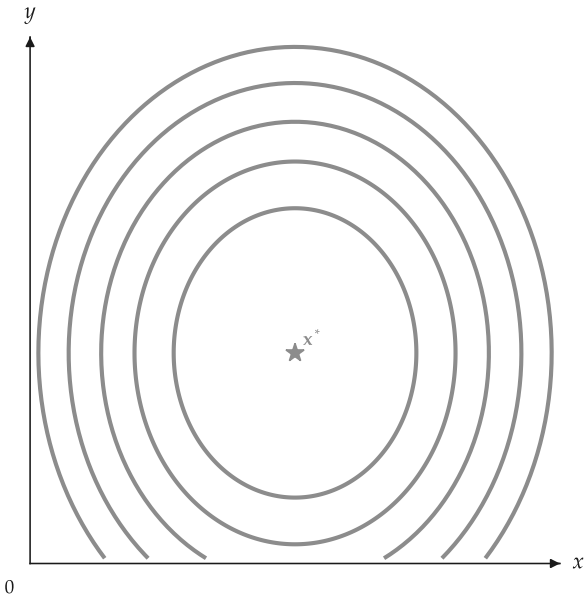


Figure 23
Satiation around the bliss point $(6, 4)$.

Example 26

```python
import numpy as np

from utility_viz import Canvas, Config, levels
from utility_viz.models import Satiation

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = Satiation(bliss_x=6.0, bliss_y=4.0)

# curves from a utility grid
X, Y = np.meshgrid(np.linspace(0.1, 12.0, 300), np.linspace(0.1, 10.0, 300))

cvs = Canvas(x_max=12, y_max=10, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.percentile(model(X, Y), n=5))  # five percentiles of u
cvs.save("satiation.png")
```

7 Advanced models

Advanced models extend the built-in families with user-defined functions or more than two goods.

Custom utility

`CustomUtility` `CustomUtility(func=⟨callable⟩, name=⟨str⟩)`

Wrap any vectorised Python callable as an utility-viz model. The callable must accept two NumPy arrays and return an array of the same shape. The example below uses

$$u(x, y) = \ln x + \ln y.$$

`func` Vectorised utility function of x and y.

`name` Display name of the model.

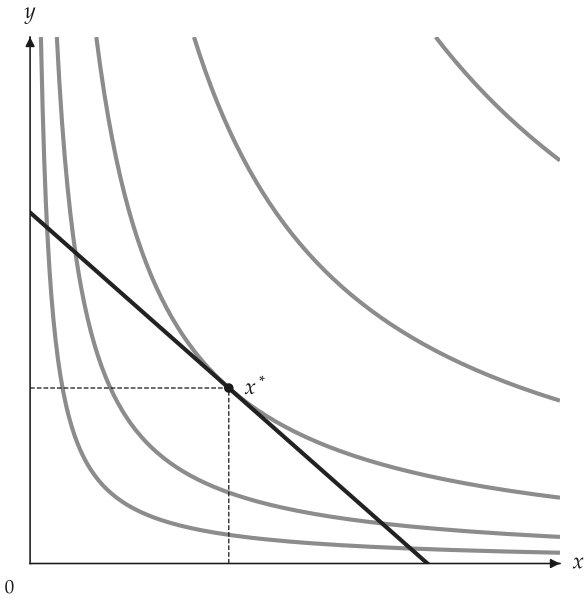


Figure 24
A custom model, $\ln x + \ln y$.

Example 27

```python
import numpy as np

from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import CustomUtility

Config.load("manual.toml").use()  # the settings behind this manual's figures

# any array function works
model = CustomUtility(func=lambda x, y: np.log(x) + np.log(y), name="log+log")
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("custom.png")
```

Many-good Cobb-Douglas

`MultiGoodCD` `MultiGoodCD(`⟨*shares*⟩`)`
`MultiGoodCD.freeze` `.freeze(`⟨*good*⟩`=`⟨*quantity*⟩`, ...)`

Cobb-Douglas preferences over N goods:

$$u(x_1, \ldots, x_N) = \prod_{i=1}^{N} x_i^{\alpha_i}.$$

`freeze()` fixes every good except x and y and returns a `CustomUtility` that can be drawn on a two-dimensional canvas.

`shares` Mapping from each good's name to its exponent α_i.
`freeze(...)` Fixed quantities of the goods other than x and y.

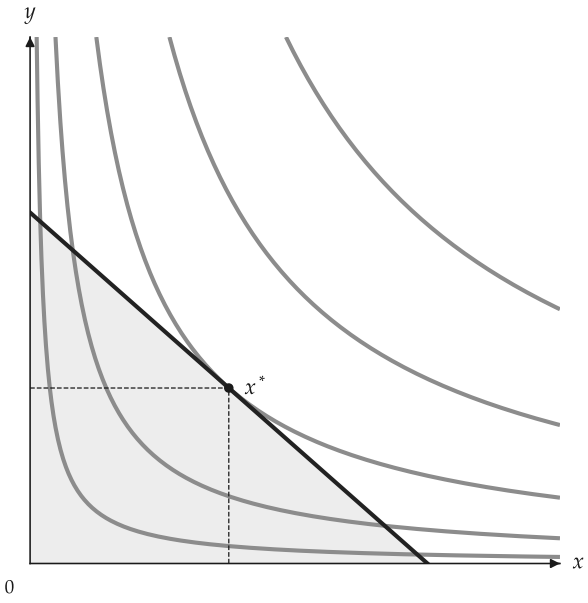


Figure 25
A three-good Cobb-Douglas model with $z = 10$ fixed.

Example 28

```
from utility_viz import Canvas, Config, levels, solve
from utility_viz.models import MultiGoodCD

Config.load("manual.toml").use()  # the settings behind this manual's figures

# three goods with z held at 10
model = MultiGoodCD({"x": 0.3, "y": 0.3, "z": 0.4})
two_good_model = model.freeze(z=10.0)
eq = solve(two_good_model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(two_good_model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("multigood.png")
```

Backend-neutral scenes

`utility_viz.core.scenes` builds the same economic components as `Canvas`, but as immutable mosaickit layers instead of drawing on matplotlib axes. Every layer has a stable id (`budget`, `budget.fill`, `equilibrium.drop`, `ic.2`, ...) and a dotted theme role (`utility.budget`, `utility.indifference.secondary`, ...), so the scene can be inspected, edited and rendered by any mosaickit backend. It is an advanced, additive API: `Canvas` and the other diagrams are unchanged.

`budget_layers` `budget_layers(⟨float⟩, ⟨float⟩, ⟨float⟩, fill=False, ...)`
`equilibrium_layers` `equilibrium_layers(⟨equilibrium⟩, drop_lines=True, ...)`
`indifference_layers` `indifference_layers(⟨model⟩, ⟨levels⟩, ⟨max⟩, ⟨max⟩, ...)`

New: v2.0.0b1

Return a tuple of layers for the budget line $p_x x + p_y y = I$, the equilibrium bundle, and the indifference curves of a model over $[0, x_{\max}] \times [0, y_{\max}]$. Styling comes from the theme roles; sparse `Stroke`, `Marker` and `Fill` overrides are accepted. `indifference_layers` returns `(layers, levels)` and traces the curves as Bézier paths with bezierkit, so it also handles non-monotone and kinked utilities.

`UTILITY_THEME` `canvas_to_tikz`

New: v2.0.0b1

`UTILITY_THEME` is a mosaickit theme with the `utility.*` roles, matching the default theme's colours. `canvas_to_tikz(canvas)` exports a scene as TikZ with native `.. controls ..` curves.

```
from mosaickit import Canvas, CanvasSpec, quadrant_axes
```

```python
from utility_viz.core.scenes import (
    UTILITY_THEME,
    budget_layers,
    canvas_to_tikz,
    equilibrium_layers,
    indifference_layers,
)
from utility_viz.models import CobbDouglas, solve

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)
u = eq.utility
ic, _ = indifference_layers(model, [0.6 * u, u, 1.4 * u], 18, 12)

spec = CanvasSpec(x_range=(0, 18), y_range=(0, 12))
canvas = (
    Canvas(spec, theme=UTILITY_THEME)
    .extend(quadrant_axes(18, 12))
    .extend(ic)
    .extend(budget_layers(2.0, 3.0, 30.0, fill=True))
    .extend(equilibrium_layers(eq))
)
canvas.save("scene.png")
tex = canvas_to_tikz(canvas)
```

The TikZ exporter handles path, fill, marker and text layers. The traced utility must be finite on the plotted region.

8 Analysis

The analysis API provides comparative statics of Marshallian demand, the Slutsky matrix, and checks for homogeneity and homotheticity of utility functions.

8.1 Comparative statics

`comparative_statics` `comparative_statics(⟨model⟩, px=⟨float⟩, py=⟨float⟩, income=⟨float⟩)`

New: v1.1.0

Estimate the six partial derivatives of two-good Marshallian demand with respect to p_x, p_y and income by finite differences, returned as a `ComparativeStatics`.

- Central finite differences around `solve(...)`, with a default relative step of `1e-3`.
- Warns when the own-price derivative is positive or the income derivative negative, the signatures of Giffen and inferior goods.

`dx_dpx` *float* $\partial x^*/\partial p_x$.

`dx_dpy` *float* $\partial x^*/\partial p_y$.

`dx_dI` *float* $\partial x^*/\partial I$.

`dy_dpx` *float* $\partial y^*/\partial p_x$.

`dy_dpy` *float* $\partial y^*/\partial p_y$.

`dy_dI` *float* $\partial y^*/\partial I$.

```python
from utility_viz import comparative_statics
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.4, beta=0.6)
cs = comparative_statics(model, px=2.0, py=3.0, income=60.0)
```

```
print(round(cs.dx_dpx, 1), round(cs.dx_dpy, 1), round(cs.dx_dI, 1))
print(round(cs.dy_dpx, 1), round(cs.dy_dpy, 1), round(cs.dy_dI, 1))

# -6.0 0.0 0.2
# 0.0 -4.0 0.2
```

8.2 Slutsky matrix

`slutsky_matrix` `slutsky_matrix(⟨model⟩, px=⟨float⟩, py=⟨float⟩, income=⟨float⟩)`

Updated: v1.2.3 The two-good Slutsky substitution matrix, computed from Marshallian demand derivatives and income effects. The result checks symmetry, negative semidefiniteness and homogeneity, and warns when a condition fails. `as_array()` returns it as a NumPy array.

```
from utility_viz import slutsky_matrix
from utility_viz.models import CobbDouglas

S = slutsky_matrix(
    CobbDouglas(alpha=0.4, beta=0.6),
    px=2.0, py=3.0, income=60.0,
)

print(round(S.s_xx, 1), round(S.s_xy, 1))
print(round(S.s_yx, 1), round(S.s_yy, 1))
print(S.as_array().round(1))

# -3.6 2.4
# 2.4 -1.6
# [[-3.6  2.4]
#  [ 2.4 -1.6]]
```

8.3 Homogeneity

`HomogeneityAnalyzer` `HomogeneityAnalyzer(⟨model⟩)`

New: v1.1.0 Check homogeneity and homotheticity of a utility function, and degree-zero homogeneity of demand.

`degree()` Estimate the degree of homogeneity; see Section 8.4.

`euler_check(x, y)` Euler-theorem residual at a bundle.

`is_homothetic()` Whether the MRS is invariant to proportional scaling.

`demand_degree_zero(px, py, income)` Whether Marshallian demand is homogeneous of degree zero.

```
from utility_viz.models import CobbDouglas, HomogeneityAnalyzer

analyzer = HomogeneityAnalyzer(CobbDouglas(alpha=0.4, beta=0.6))
result = analyzer.degree()

print(round(result.degree, 6))
print(result.returns_to_scale)
print(round(analyzer.euler_check(3.0, 4.0), 6))
print(analyzer.is_homothetic())
print(analyzer.demand_degree_zero(px=2.0, py=3.0, income=60.0))

# 1.0
# ReturnsToScale.CONSTANT
# 0.0
# True
# True
```

8.4 Returns to scale

`degree()` returns a `HomogeneityResult` with these fields:

`degree` *float* — Estimated degree of homogeneity.

`is_homogeneous` *bool* — Whether the function is homogeneous.

`returns_to_scale` *ReturnsToScale* — Returns-to-scale class, one of the values below.

For Cobb-Douglas utility the degree is $\alpha + \beta$:

$$u(\lambda x, \lambda y) = \lambda^{\alpha+\beta} u(x, y).$$

`ReturnsToScale` — `ReturnsToScale.INCREASING`, `CONSTANT`, `DECREASING`, `NOT_HOMOGENEOUS`

The returns-to-scale class of a function.

`INCREASING` *ReturnsToScale* — $\alpha + \beta > 1$, increasing returns.

`CONSTANT` *ReturnsToScale* — $\alpha + \beta = 1$, constant returns.

`DECREASING` *ReturnsToScale* — $\alpha + \beta < 1$, decreasing returns.

`NOT_HOMOGENEOUS` *ReturnsToScale* — No consistent degree of homogeneity.

```
def shifted_utility(x, y):
    return x**0.4 * y**0.6 + 1.0


models = [
    CobbDouglas(alpha=0.7, beta=0.6),
    CobbDouglas(alpha=0.4, beta=0.6),
    CobbDouglas(alpha=0.2, beta=0.5),
    shifted_utility,
]

for model in models:
    result = HomogeneityAnalyzer(model).degree()
    degree = None if result.degree is None else round(result.degree, 1)
    print(degree, result.returns_to_scale.name)

# 1.3 INCREASING
# 1.0 CONSTANT
# 0.7 DECREASING
# None NOT_HOMOGENEOUS
```

9 Themes and configuration

A `Theme` defines the default look of a diagram; a `Config` loads theme overrides and font settings from a TOML file. `Canvas`, `Figure`, `DemandDiagram`, price-effect decompositions and `EdgeworthBox` share these style objects.

9.1 Built-in themes

`themes`

```
from utility_viz import Canvas, themes

cvs = Canvas(theme=themes.paper)
```

Updated: v1.11.0

The Python API provides the themes below. Theme objects are immutable; to change a field, create a new `Theme` or override the field with a `Config`.

Table 6
Built-in themes

Name	Character
`default`	Colour-blind-friendly default palette
`colorblind`	Alias with the same palette as `default`
`nord`	Cool theme on the Nord palette
`paper`	Transparent background and fine lines for print
`monochrome`	Black and white; layers told apart by line style and marker
`presentation`	Larger labels, lines and markers for projection
`dark`	Dark background with matching foreground colours

The Python API and the `base` key of a configuration file accept all seven names[5].

The default theme takes its colours from a colour-blind-friendly palette (thriveth, 2014). `themes.COLORBLIND_CYCLE_HEX` and `themes.COLORBLIND_CYCLE_RGB` give the palette as hexadecimal and RGB values. Economics teaching leans heavily on diagrams, so meaning should never rest on colour alone, or visually impaired students cannot follow it (Kugler & Andrews, 1996).

The figures below show the same Cobb-Douglas diagram in each built-in theme except `dark`, with the manual's line weights on a white page.

Example 29

```
from dataclasses import replace

from utility_viz import Canvas, levels, solve, themes
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

# theme colours with manual line weights
theme = replace(
    themes.default,
    background_color="#ffffff",
    ic_linewidth=2.6,
    budget_linewidth=2.6,
    eq_markersize=5.0,
)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$", theme=theme)
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("theme_default.png")
```

[5]Passed directly to `--theme` on the command line, only the six themes other than the `colorblind` alias are accepted.

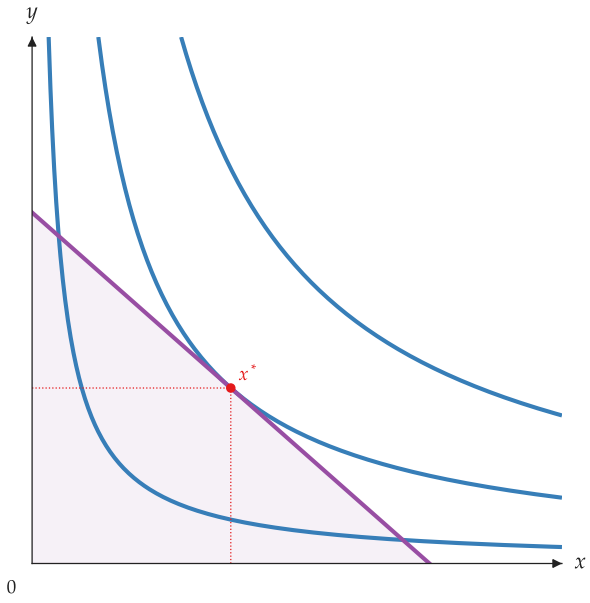


Figure 26
The `default` theme.

Example 30

```python
from dataclasses import replace

from utility_viz import Canvas, levels, solve, themes
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

# theme colours with manual line weights
theme = replace(
    themes.nord,
    background_color="#ffffff",
    ic_linewidth=2.6,
    budget_linewidth=2.6,
    eq_markersize=5.0,
)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$", theme=theme)
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("theme_nord.png")
```

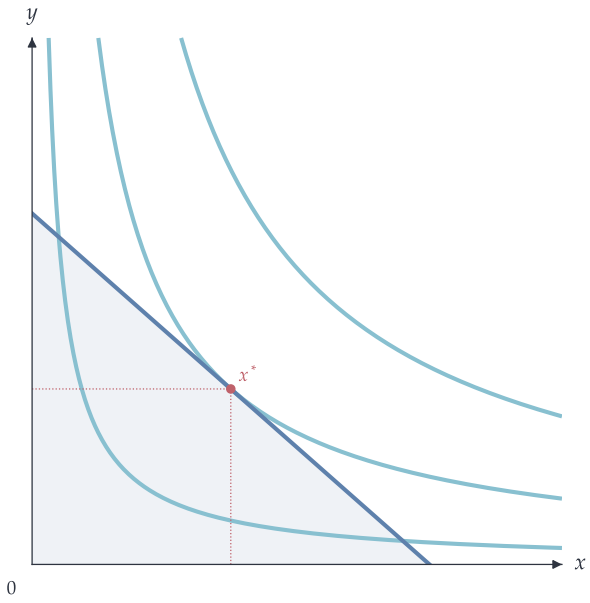


Figure 27
The `nord` theme.

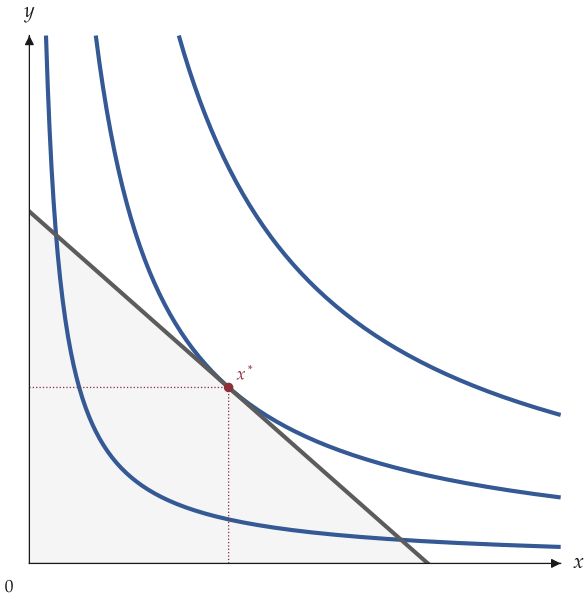


Figure 28
The paper theme.

Example 31

```python
from dataclasses import replace

from utility_viz import Canvas, levels, solve, themes
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

# theme colours with manual line weights
theme = replace(
    themes.paper,
    background_color="#ffffff",
    ic_linewidth=2.6,
    budget_linewidth=2.6,
    eq_markersize=5.0,
)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$", theme=theme)
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("theme_paper.png")
```

Example 32

```python
from dataclasses import replace

from utility_viz import Canvas, levels, solve, themes
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

# theme colours with manual line weights
theme = replace(
    themes.monochrome,
    background_color="#ffffff",
    ic_linewidth=2.6,
    budget_linewidth=2.6,
    eq_markersize=5.0,
)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$", theme=theme)
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("theme_monochrome.png")
```

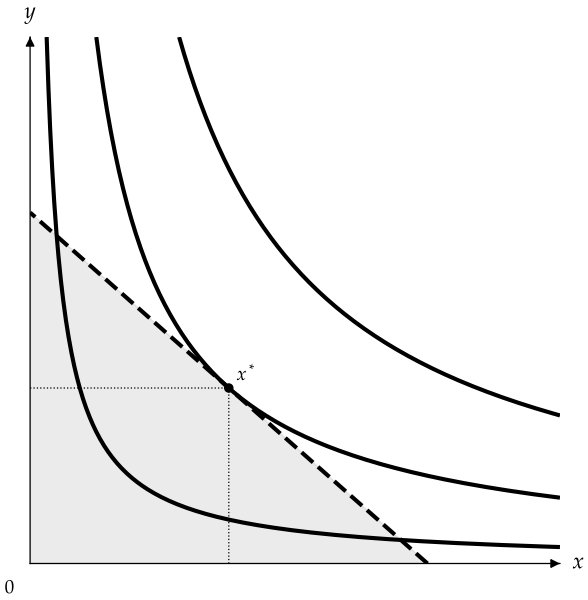


Figure 29
The `monochrome` theme.

Example 33

```python
from dataclasses import replace

from utility_viz import Canvas, levels, solve, themes
from utility_viz.models import CobbDouglas

model = CobbDouglas(alpha=0.5, beta=0.5)
eq = solve(model, px=2.0, py=3.0, income=30.0)

# theme colours with manual line weights
theme = replace(
    themes.presentation,
    background_color="#ffffff",
    ic_linewidth=2.6,
    budget_linewidth=2.6,
    eq_markersize=5.0,
)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$", theme=theme)
cvs.add_utility(model, levels=levels.around(eq.utility, n=3))
cvs.add_budget(2.0, 3.0, 30.0, fill=True)
cvs.add_equilibrium(eq)
cvs.save("theme_presentation.png")
```

9.2 Custom themes

`Theme`

```python
from utility_viz import Theme

my_theme = Theme(
    name="custom",
    background_color="white",
    label_scale=1.0,
    axis_color="#333333",
    label_color="#333333",
    ic_color="#2563eb",
    secondary_ic_color="#93c5fd",
    secondary_ic_opacity=0.45,
    budget_color="#dc2626",
    eq_color="#16a34a",
)
```

Updated: v1.12.0

A `Theme` holds default colours, widths and styles for diagram elements. Fields not given keep the class defaults; style arguments passed explicitly to a drawing method take precedence over the theme.

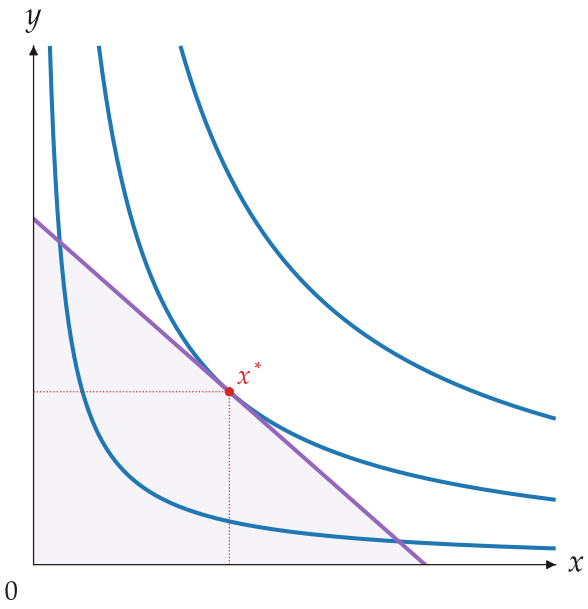


Figure 30
The `presentation` theme.

`name` — *str*
Theme identifier.

`background_color` — *str | None* · default `None`
Background of the figure and axes; `None` keeps it transparent.

`label_scale` — *float* · default `1.0`
Multiplier for ordinary label sizes.

`axis_color` — *str* · default `"#222222"`
Axis colour.

`label_color` — *str* · default `"#222222"`
Colour of axis labels and origin text.

`ic_color` — *str* · default `"#377EB8"`
Indifference-curve colour.

`ic_linewidth` — *float* · default `1.8`
Indifference-curve width.

`secondary_ic_color` — *str | None* · default `None`
Colour of non-focal indifference curves; `None` uses `ic_color`.

`secondary_ic_linewidth` — *float* · default `1.0`
Width of non-focal indifference curves.

`secondary_ic_opacity` — *float* · default `0.45`
Opacity of non-focal indifference curves.

`path_color` — *str* · default `"#4DAF4A"`
Colour of PCC and ICC paths.

`budget_color` — *str* · default `"#984EA3"`
Budget-line colour.

`budget_fill_alpha` — *float* · default `0.08`
Opacity of the budget-set shading.

`eq_color` — *str* · default `"#E41A1C"`
Colour of the equilibrium and its drop lines.

`eq_markersize` — *float* · default `4.0`
Equilibrium marker size.

`Theme` also holds colours and widths for rays, kinks, compensated budget lines and price-effect arrows. `axis_stroke`, `drop_stroke`, `projection_stroke`, `guide_stroke` and `box_stroke` hold the defaults of other lines; markers, labels, fills and legends come from `Marker`, `Label`, `Fill` and `Legend` properties (Section 4.3).

9.3 Configuration files

Config.load
Config.use
Config.reset

```
from utility_viz import Config

Config.load("utility-viz.toml").use()
# later diagrams use these settings

Config.reset()
# back to the built-in defaults
```

New: v1.10.0 `Config.load()` reads a TOML file and returns a `Config`; `use()` makes it the default for later diagrams, and `reset()` restores the built-in defaults.

Tutorial: one shared style

Figures in one set of lecture notes or one paper usually share their colours and line styles. Rather than repeating the same arguments on every `Canvas`, write the style once in an `utility-viz.toml` at the project root, where both Python and the command line read it.

1. **Create a template.** Run the following in the project root. It writes a commented `utility-viz.toml` listing the names and fields each section accepts:
   ```
   utility-viz init
   ```
   Every setting in the template starts commented out; keep only what you change and delete the rest.
2. **Pick a base theme.** `base` names the built-in theme to start from (Table 6); everything else overrides it:
   ```
   base = "paper"
   ```
3. **Override colours and styles.** Write only the fields you change:
   ```
   [color]
   ic = "#1B4F72"
   eq = "#C0392B"

   [stroke.budget]
   width = 1.6
   style = "dashed"

   [marker.eq]
   size = 5

   [label.point]
   fontsize = 11
   position = "right"
   ```
   Section names map to theme properties: `ic` under `[color]` sets `ic_color`, and `[stroke.budget]` sets `theme.budget_stroke`. The fields inside a section are the arguments of `Stroke`, `Marker`, `Label`, `Fill` or `Legend` (Section 4.3). Fonts go under `[font]` as `text` and `math`. Fields you leave out keep the base theme's values.
4. **Use it from Python.** Load it once, before creating any diagram:
   ```
   from utility_viz import Config

   Config.load().use()   # reads ./utility-viz.toml by default
   ```
   Every `Canvas`, `Figure`, `DemandDiagram` and `EdgeworthBox` created afterwards uses these settings; `Config.reset()` restores the defaults.
5. **Use it from the command line.** `utility-viz plot` reads `utility-viz.toml` from the current directory on its own; pass another file with `--config`:
   ```
   utility-viz plot --config utility-viz.toml --model cobb-douglas \
     --px 2 --py 3 --income 30 --output figure.png
   ```

Settings apply in this order, highest first: arguments passed to `Canvas` or a drawing method, then the configuration file, then the base theme. For example, `Canvas(theme=themes.default)` ignores the file's theme.

A mistake in the file makes `Config.load()` raise `InvalidParameterError` with the valid names. For instance, writing `[stroke.budgt]` for `[stroke.budget]` gives:

```
utility-viz.toml: [stroke.budgt]: no such setting (choose: axis, box,
budget, ...)
```

This manual's settings

Every example in this manual loads `manual.toml` first. It sets the look of the figures: grey curves, black axes and guides, black points and a white page. It leaves the font out so that it runs anywhere; the figures use TeX Gyre Pagella, and without that font you get the default typeface.

```
# Settings for every manual figure
# Grey curves and black axes on white
# Load them before drawing
#
#     Config.load("manual.toml").use()
#
# Sections map to Theme properties

base = "default"

# Font is TeX Gyre Pagella but unset
# Add [font] text = "TeX Gyre Pagella" to use it

[color]
background = "#ffffff"
axis = "#1a1a1a"
label = "#1a1a1a"
ic = "#8c8c8c"              # indifference curves
secondary_ic = "#b8b8b8"    # the curves beside a highlighted one
path = "#1a1a1a"            # price- and income-consumption paths
budget = "#222222"
eq = "#1a1a1a"              # equilibrium and other points
ray = "#8c8c8c"
kink = "#8c8c8c"
sub_effect = "#8c8c8c"
inc_effect = "#8c8c8c"
compensated_budget = "#1a1a1a"
subsistence = "#8c8c8c"
contract = "#8c8c8c"
core = "#1a1a1a"
price = "#8c8c8c"
walrasian = "#1a1a1a"

[stroke.axis]
width = 1.0
style = "solid"
arrow = "-|>"

[stroke.ic]
width = 2.6

[stroke.secondary_ic]
width = 1.8
opacity = 1.0

[stroke.path]
width = 1.4

[stroke.budget]
width = 2.6

[stroke.ray]
width = 0.9

[stroke.substitution]
width = 1.8
```

```toml
[stroke.income]
width = 1.8

[stroke.compensated_budget]
width = 1.0

[stroke.contract]
width = 2.6

[stroke.price]
width = 2.6

[stroke.drop]               # guides from a point to the axes
width = 0.8
style = "dashed"
color = "#1a1a1a"

[stroke.projection]
width = 0.8
style = "dashed"
color = "#1a1a1a"

[stroke.guide]
width = 0.8
style = "dashed"
color = "#1a1a1a"

[marker.eq]
size = 5

[marker.point]
size = 5

[label.point]               # text beside a point such as x*
fontsize = 14
position = "right"
offset = 10

[fill.budget]
opacity = 0.08
```

File lookup

`Config.discover` `Config.discover([⟨path⟩], directory=None)`

New: v2.0.0b1 Find and read a configuration file without failing when there is none.

`Config.load()`, `Config.discover()` and `utility-viz plot` resolve the file in this order:

1. the path given explicitly (`Config.load("file.toml")` or `--config`), which must exist and is the only file read;
2. `utility-viz.toml` in the current directory (or `directory=`);
3. the legacy `econ-viz.toml`, with a deprecation warning;
4. the built-in defaults.

When both files exist, `utility-viz.toml` wins and the legacy one is ignored with a warning. The two functions differ only when no file is found: `Config.load()` raises `InvalidParameterError` ("config file not found"), as in 1.x, while `Config.discover()` returns the built-in defaults. `utility-viz init --migrate` turns a legacy file into `utility-viz.toml` (Section 11.3).

10 Export

10.1 Output formats

Canvas.save `cvs.save(⟨path⟩)`

Write the figure to `path`; the format follows the extension. `Figure`, `DemandDiagram` and `EdgeworthBox` have the same method.

- PNG: raster, at the canvas `dpi` (default 300).
- PDF, SVG: vector output that stays sharp when printed or scaled.
- TeX: TikZ code (`.tex`) for inclusion in LaTeX documents; adjust it with the keyword arguments `tikz_scale` and `tikz_standalone`.

```
cvs.save("figure.png")   # PNG with canvas DPI default 300
cvs.save("figure.pdf")   # PDF vector
cvs.save("figure.svg")   # SVG vector
cvs.save("figure.tex")   # TikZ
```

The `dpi` parameter of `Canvas` only affects raster output; lower it for faster previews:

```
cvs = Canvas(x_max=20, y_max=15, dpi=150)   # lower DPI for a faster
preview
```

10.2 Animated GIFs

Export a GIF with `Animator.save()`; Section 12 has complete examples.

```
from utility_viz.core.animation import Animator

Animator(draw_frame, frames=frames).save("animation.gif", fps=12, dpi=120)
```

10.3 Interactive window

`cvs.show()` opens a live matplotlib window instead of saving. In the CLI, omitting `--output` has the same effect (Section 11.2):

```
utility-viz plot --model cobb-douglas --px 2 --py 3 --income 30
```

11 Command-line interface

The command-line interface plots diagrams, runs batches and prints demand formulas. Install it as a standalone tool with `uv tool install utility-viz`, or prefix each command with `uv run` inside a uv project (Section 2.5). The 1.x command `econ-viz`, installed by the econ-viz compatibility distribution, still works in 2.x: it warns that it is deprecated and runs `utility-viz` with the same arguments (Section 2.8).

Table 7 CLI commands

Command	Description
`utility-viz help [⟨command⟩]`	Show help for the CLI or a specific command
`utility-viz models`	List all supported utility models
`utility-viz plot ...`	Generate and export a diagram
`utility-viz solve-tex ...`	Print a closed-form Marshallian demand as TeX
`utility-viz init [path]`	Create an `utility-viz.toml` configuration template

11.1 Help and models

utility-viz help

`utility-viz help [⟨command⟩]`

Without an argument, list all commands; with one, show its options.

```
utility-viz help          # all commands
utility-viz help plot     # plot options
utility-viz help models   # models options
```

utility-viz models

List the model names and parameters the CLI supports. `--model` takes kebab-case names, which differ from the Python class names (Table 8). The parameter options are described in Section 11.2.

Table 8 CLI models

Model	`--model` value	Required	Optional
Cobb-Douglas	`cobb-douglas`	`alpha`, `beta`	–
CES	`ces`	`rho`	`alpha`, `beta`
Perfect complements	`leontief`	`a`, `b`	–
Perfect substitutes	`perfect-substitutes`	`a`, `b`	–
Satiation	`satiation`	`bliss-x`, `bliss-y`	`a`, `b`
Quasi-linear	`quasi-linear`	–	`v-func`, `linear-in`
Stone-Geary	`stone-geary`	–	`alpha`, `beta`, `bar-x`, `bar-y`
Translog	`translog`	–	`alpha-0`, `alpha-x`, `alpha-y`, `beta-xx`, `beta-yy`, `beta-xy`

11.2 Plotting

utility-viz plot

`utility-viz plot --model ⟨name⟩ ⟨options⟩`
`utility-viz plot --latex ⟨expression⟩ ⟨options⟩`

Give the model with `--model`, or the utility function with `--latex`, not both. With `--output` the diagram is written to a file; without it an interactive window opens.

The budget line and equilibrium need both prices and income. `--no-budget`, `--no-equilibrium` and `--no-curves` each omit one layer and can be combined (Table 9).

Table 9 Layer options

Case	Result
`--px`, `--py` and `--income` all given	Budget line drawn and equilibrium solved
Any price or income missing	Indifference curves only, no budget line or equilibrium
`--no-equilibrium` added	No solve and no equilibrium, even with prices and income
`--no-budget` added	Budget line hidden, even with prices and income
`--no-curves` with prices and income	Budget line and equilibrium only, no indifference curves

```
# named model
utility-viz plot --model cobb-douglas --alpha 0.5 --beta 0.5 ...

# LaTeX expression
utility-viz plot --latex "x^{0.4} y^{0.6}" ...
```

Example 34

```
# Cobb-Douglas, shaded budget set
utility-viz plot --model cobb-douglas --alpha 0.5 --beta 0.5 \
             --px 2 --py 3 --income 30 \
             --fill --output cobb_douglas.png

# LaTeX input, Nord theme, expansion path
utility-viz plot --latex "x^{0.4} y^{0.6}" \
             --px 2 --py 3 --income 30 \
             --theme nord --show-ray \
             --output cd_latex.png

# perfect complements, larger canvas
utility-viz plot --model leontief --a 1 --b 2 \
             --px 2 --py 3 --income 30 \
             --x-max 20 --y-max 15 \
             --output leontief.png

# CES, curves only
utility-viz plot --model ces --rho -0.5 \
             --x-max 20 --y-max 15 --n-curves 6 \
             --no-budget --no-equilibrium \
             --output ces.png

# satiation (bliss point)
utility-viz plot --model satiation --bliss-x 6 --bliss-y 4 \
             --x-max 12 --y-max 10 \
             --no-budget --no-equilibrium \
             --output satiation.png

# no --output: open a window
utility-viz plot --model cobb-douglas --px 2 --py 3 --income 30
```

Model selection

`--model, -m` Model name; `utility-viz models` lists them all.

`--latex, -l` LaTeX expression for Cobb-Douglas, perfect complements, perfect substitutes or CES (Section 14).

Prices and income

`--px` Price of good x.

`--py` Price of good y.

`--income` Consumer income.

Model parameters

`--alpha` Alpha parameter (Cobb-Douglas, CES). default `0.5`

`--beta` Beta parameter (Cobb-Douglas, CES). default `0.5`

`--a` Parameter a (perfect complements, perfect substitutes, satiation). default `1.0`

`--b` Parameter b (perfect complements, perfect substitutes, satiation). default `1.0`

`--rho` Substitution parameter (CES). default `0.5`

`--bliss-x` Bliss-point x-coordinate (satiation). default `5.0`

`--bliss-y` Bliss-point y-coordinate (satiation). default `5.0`

Quasi-linear, Stone-Geary and Translog take these further options:

`--v-func` Non-linear function of the quasi-linear model, `log` or `sqrt`.
default `log`

`--linear-in` Good that enters the quasi-linear model linearly, `x` or `y`.
default `y`

`--bar-x` Stone-Geary subsistence quantity $\overline{x}$.
default `1.0`

`--bar-y` Stone-Geary subsistence quantity $\overline{y}$.
default `1.0`

`--alpha-0` Translog intercept.
default `0.0`

`--alpha-x` Translog first-order weight on $\ln x$.
default `0.5`

`--alpha-y` Translog first-order weight on $\ln y$.
default `0.5`

`--beta-xx` Translog curvature in x.
default `0.0`

`--beta-yy` Translog curvature in y.
default `0.0`

`--beta-xy` Translog interaction between x and y.
default `0.0`

```
utility-viz plot --model stone-geary --bar-x 2 --bar-y 2 \
             --px 2 --py 3 --income 30 \
             --x-max 20 --y-max 15 --output stone_geary.png
```

Subsistence spending here is $2 \times 2 + 3 \times 2 = 10$, below the income of 30. With an income of 10 or less, the solve fails because the budget lies outside the model's domain.

Canvas and layers

`--x-max` Horizontal axis limit.
default `10`

`--y-max` Vertical axis limit.
default `10`

`--x-label` Horizontal axis label.
default `x`

`--y-label` Vertical axis label.
default `y`

`--title` Figure title.

`--theme` Theme name: `default`, `nord`, `paper`, `monochrome`, `presentation` or `dark` (Section 9).
default `default`

`--config` Path to a TOML configuration file. Command-line options take precedence over the file. Without it, `utility-viz.toml` (or the legacy `econ-viz.toml`) in the current directory is used if present (Section 9.3.3).

`--n-curves` Number of indifference curves.
default `5`

`--dpi` Raster output resolution.
default `300`

`--fill` Shade the feasible set below the budget line.

`--show-ray` Draw the expansion-path ray through the optimum.

`--no-budget` Omit the budget line.

`--no-equilibrium` Omit the equilibrium point.

`--no-curves` Omit the indifference curves.

`--output, -o` Output file (`.png`, `.pdf`, `.svg`, `.tex`); omit it to open an interactive window.

For batch runs, give each parameter set its own file name and create the output directory first. `--x-max` and `--y-max` do not follow the prices; when an intercept or the optimum falls outside the canvas, raise the limits as well.

11.3 Creating a configuration file

`utility-viz init` `utility-viz init [⟨path⟩] [--force] [--migrate]`

New: v1.10.0 Write a commented `utility-viz.toml` template. Without a path it goes in the current directory; an existing file is overwritten only with `--force`. `--migrate` instead creates `utility-viz.toml` from a legacy `econ-viz.toml` in the same directory and keeps the old file.

```
utility-viz init
utility-viz init config/figures.toml
utility-viz init --migrate
utility-viz plot --config utility-viz.toml --model cobb-douglas \
             --px 2 --py 3 --income 30 --output figure.png
```

11.4 Demand formulas

`utility-viz solve-tex` `utility-viz solve-tex --model ⟨name⟩ ⟨options⟩`

Print the closed-form Marshallian demand as TeX, for any document that supports TeX math.

```
# numeric parameters
utility-viz solve-tex --model cobb-douglas --alpha 0.4 --beta 0.6

# symbolic parameters
utility-viz solve-tex --model cobb-douglas --symbolic-params

# custom price and income symbols
utility-viz solve-tex --model leontief --a 2 --b 3 \
                  --px-symbol p_1 --py-symbol p_2 --income-symbol M
```

Supports `cobb-douglas`, `leontief`, `perfect-substitutes` and their LaTeX expressions. The command prints the formula only and writes no figure.

`--symbolic-params` Keep model parameters as symbols, such as α and β; without it the given values are substituted.

`--px-symbol` default `p_x` Symbol for the price of x.

`--py-symbol` default `p_y` Symbol for the price of y.

`--income-symbol` default `I` Symbol for income.

12 Animation

`Animator` calls a drawing function over a sequence of parameter values and exports the returned `Canvas` or `Figure` objects as a GIF. It needs the `animation` extra (Section 2.4). Import it from `utility_viz.core.animation`; it was `econ_viz.animation` in 1.x.

`Animator` `Animator(⟨draw⟩, frames=⟨values⟩).save(⟨path⟩, fps=12, dpi=120, loop=0)`

New: v1.4.0 Call `draw(value)` for every value in `frames` and write the canvases as the frames of a GIF.

- `save()` uses Pillow, so no `ffmpeg` dependency is required.
- Frames are composited onto white before export, which prevents frame-stacking artefacts.

- `fps`, `dpi` and `loop` set the frame rate, resolution and loop count.

Example 35

```python
import numpy as np

from utility_viz import Canvas, levels, solve
from utility_viz.core.animation import Animator
from utility_viz.models import CobbDouglas

def draw(px: float) -> Canvas:
    model = CobbDouglas(alpha=0.5, beta=0.5)
    eq = solve(model, px=px, py=2.0, income=20.0)
    lvls = levels.around(eq.utility, n=5)

    return (
        Canvas(
            x_max=14, y_max=12,
            x_label="X_1", y_label="X_2",
            title="Price sweep"
        )
        .add_utility(model, levels=lvls)
        .add_budget(px=px, py=2.0, income=20.0, fill=True)
        .add_equilibrium(eq, show_ray=True, drop_dashes=True)
    )

Animator(draw, frames=np.linspace(1.0, 6.0, 45)).save(
    "price_sweep.gif",
    fps=12,
    dpi=120,
)
```

`examples/scripts/animation.py` provides four examples[6]:

- Parameter sweep: prices and income fixed, a utility-function parameter varies.
- Price sweep: utility function and p_y fixed, p_x varies and the budget line rotates, as in the example above.
- Income sweep: utility function and prices fixed, income varies and the budget line shifts.
- Budget-only sweep: the budget line alone, without the utility layers.

The four differ only in which values `draw()` holds fixed and which one it varies. For the parameter sweep, let the `draw()` above take `alpha` and fix the price at $p_x = 2$:

```python
def draw(alpha: float) -> Canvas:
    model = CobbDouglas(alpha=alpha, beta=1.0 - alpha)
    eq = solve(model, px=2.0, py=2.0, income=20.0)
    ...

Animator(draw, frames=np.linspace(0.1, 0.9,
45)).save("parameter_sweep.gif")
```

For the income sweep, `draw()` takes `income` and passes it to `add_budget()` and `solve()`. For the budget line alone, leave out `add_utility()` and `add_equilibrium()`.

Figure 31 to Figure 33 each show four frames.

[6] Each example covers Cobb-Douglas, CES, perfect substitutes and perfect complements.

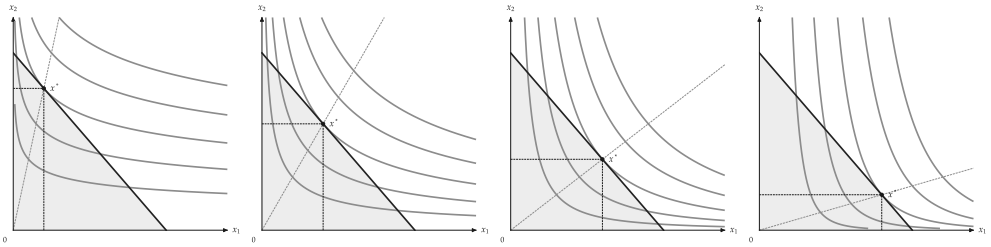


Figure 31
Parameter sweep: α varies with $\beta = 1 - \alpha$.

Example 36

```python
import numpy as np

from utility_viz import Canvas, Config, levels, solve
from utility_viz.core.animation import Animator
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures


def draw(alpha: float) -> Canvas:
    """One frame: the weight on x is alpha, the weight on y is 1 - alpha."""
    model = CobbDouglas(alpha=alpha, beta=1.0 - alpha)
    eq = solve(model, px=2.0, py=2.0, income=20.0)
    cvs = Canvas(x_max=14, y_max=12, x_label="$x_1$", y_label="$x_2$")
    cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
    cvs.add_budget(2.0, 2.0, 20.0, fill=True)
    cvs.add_equilibrium(eq, show_ray=True)
    return cvs


Animator(draw, frames=np.linspace(0.1, 0.9, 45)).save("parameter_sweep.gif", fps=12,
dpi=120)
```

Each figure below shows four frames of its GIF, which has 45.

Example 37

```python
import numpy as np

from utility_viz import Canvas, Config, levels, solve
from utility_viz.core.animation import Animator
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
# same curves in every frame
fixed = levels.around(solve(model, px=2.0, py=2.0, income=20.0).utility, n=5)


def draw(px: float) -> Canvas:
    """One frame: the price of x is px."""
    eq = solve(model, px=px, py=2.0, income=20.0)
    cvs = Canvas(x_max=14, y_max=12, x_label="$x_1$", y_label="$x_2$")
    cvs.add_utility(model, levels=fixed)
    cvs.add_budget(px, 2.0, 20.0, fill=True)
    cvs.add_equilibrium(eq, show_ray=True)
    return cvs


Animator(draw, frames=np.linspace(1.0, 6.0, 45)).save("price_sweep.gif", fps=12, dpi=120)
```

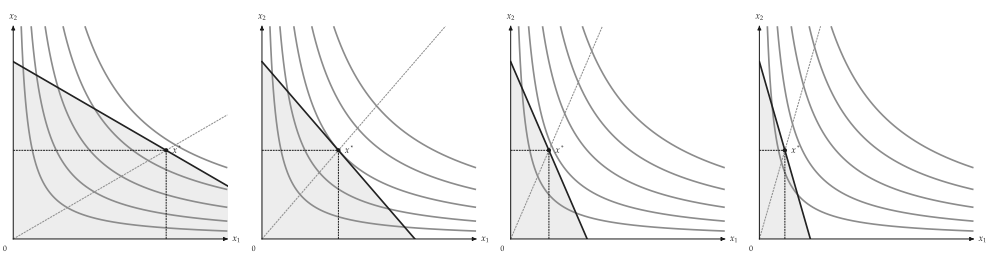


Figure 32
Price sweep with $p_y = 2$ and $I = 20$ fixed.

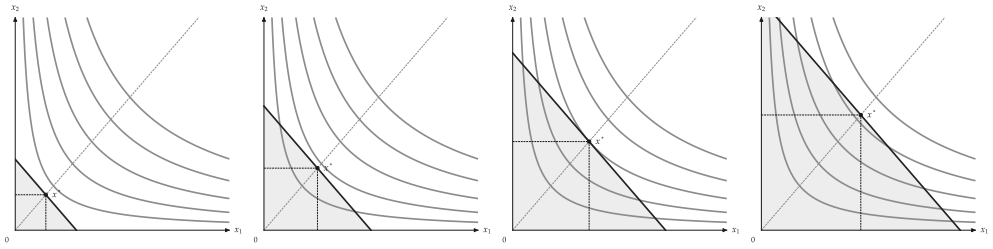


Figure 33
Income sweep with $p_x = p_y = 2$ fixed.

Example 38

```python
import numpy as np

from utility_viz import Canvas, Config, levels, solve
from utility_viz.core.animation import Animator
from utility_viz.models import CobbDouglas

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = CobbDouglas(alpha=0.5, beta=0.5)
# same curves in every frame
fixed = levels.around(solve(model, px=2.0, py=2.0, income=20.0).utility, n=5)


def draw(income: float) -> Canvas:
    """One frame: income is `income`."""
    eq = solve(model, px=2.0, py=2.0, income=income)
    cvs = Canvas(x_max=14, y_max=12, x_label="$x_1$", y_label="$x_2$")
    cvs.add_utility(model, levels=fixed)
    cvs.add_budget(2.0, 2.0, income, fill=True)
    cvs.add_equilibrium(eq, show_ray=True)
    return cvs


Animator(draw, frames=np.linspace(8.0, 36.0, 45)).save("income_sweep.gif", fps=12, dpi=120)
```

13 Interactive widgets

`WidgetViewer` provides sliders and numeric input boxes in Jupyter and redraws the figure when a parameter changes. It needs the `interactive` extra (Section 2.4). Import it from `utility_viz.core.interactive`; it was `econ_viz.interactive` in 1.x.

`WidgetViewer` `WidgetViewer(`⟨*draw*⟩`, `⟨*name*⟩`=(`⟨*min*⟩`, `⟨*max*⟩`, `⟨*step*⟩`), ...).show()`

New: v1.4.0

Pass a drawing function `draw` and give each parameter's range as `(min, max, step)`. Each parameter gets a linked `FloatSlider` and `FloatText`; when a value changes, `draw` is called again and its figure replaces the old one in the cell.

`draw` follows the same contract as for `Animator` (Section 12), but may take several parameters. Starting from the price-sweep example there, let `draw()` also take `alpha`:

Example 39

```python
from utility_viz.core.interactive import WidgetViewer

def draw(alpha: float, px: float) -> Canvas:
    model = CobbDouglas(alpha=alpha, beta=1.0 - alpha)
    ...  # rest as in the animation example

WidgetViewer(
    draw,
    alpha=(0.2, 0.8, 0.05),
    px=(1.0, 6.0, 0.25),
).show()
```

13.1 Notebook setup

In a fresh Jupyter or Colab runtime, run the steps in this order[7]:

1. Run the install cell.
2. If the install upgraded ipywidgets, traitlets or IPython, restart the runtime.
3. After the restart, continue from the import cell without reinstalling.

13.2 Widgets or GIFs

Use `WidgetViewer` when users should adjust the parameters themselves; use `Animator` to export a GIF when a fixed sweep should play in slides or on a web page (Section 12).

14 LaTeX parsing

`parse_latex` `parse_latex(`⟨*expression*⟩`)`

Parse a LaTeX utility function into a model, usable with `solve()`, `Canvas` or any other API that takes a model.

```
from utility_viz import parse_latex

model = parse_latex(r"x^{0.4} y^{0.6}")  # CobbDouglas(alpha=0.4,
beta=0.6)
model = parse_latex(r"\min(2x, 3y)")     # Leontief(a=2.0, b=3.0)
model = parse_latex(r"2x + 3y")          # PerfectSubstitutes(a=2.0,
b=3.0)
```

14.1 Supported forms

Table 10 lists the accepted forms. Omitted coefficients and exponents default to 1.

Table 10 Utility functions recognised by `parse_latex`.

Family	Pattern	Example
Cobb-Douglas	`x^{α} y^{β}` or `x^α y^β`	`x^{0.3} y^{0.7}`
Perfect complements	`\min(ax, by)` or `min(ax, by)`	`\min(2x, y)`
Perfect substitutes	`ax + by`	`3x + 1.5y`
CES	`(αx^ρ + βy^ρ)^{1/ρ}`	`(0.4x^{-0.5}+0.6y^{-0.5})^{-2}`

A leading `U =` or `u(x, y) =` may be omitted:

```
U(x,y) = x^{0.5} y^{0.5}
U = x^{0.5} y^{0.5}
x^{0.5} y^{0.5}
```

[7] The Playground notebook in the utility-viz repository, `notebook/econ-viz Playground.ipynb` (which keeps its historical name), follows this flow; open it in Jupyter, VS Code or Colab. It does not reinstall when utility-viz is already present.

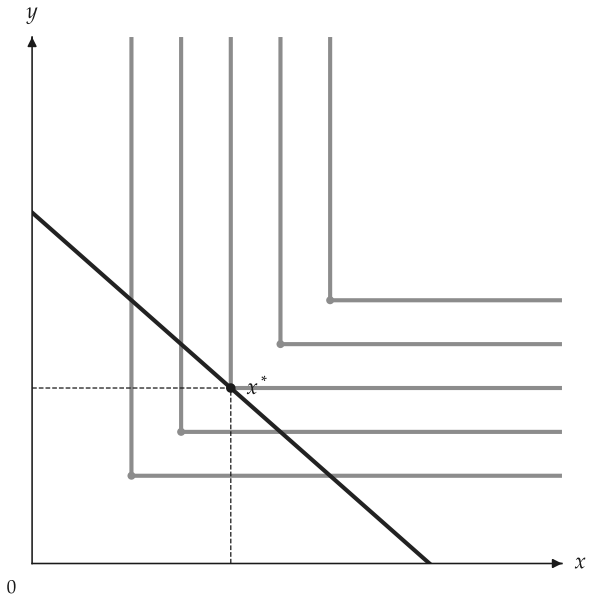


Figure 34 `parse_latex(r"\min(2x, 3y)")`.

Example 40

```
from utility_viz import Canvas, Config, levels, parse_latex, solve

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = parse_latex(r"\min(2x, 3y)")  # Leontief(a=2.0, b=3.0)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5), show_kinks=True)
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("leontief.png")
```

Example 41

```
from utility_viz import Canvas, Config, levels, parse_latex, solve

Config.load("manual.toml").use()  # the settings behind this manual's figures

model = parse_latex(r"2x + 3y")  # PerfectSubstitutes(a=2.0, b=3.0)
eq = solve(model, px=2.0, py=3.0, income=30.0)

cvs = Canvas(x_max=20, y_max=15, x_label="$x$", y_label="$y$")
cvs.add_utility(model, levels=levels.around(eq.utility, n=5))
cvs.add_budget(2.0, 3.0, 30.0)
cvs.add_equilibrium(eq)
cvs.save("perfect_substitutes.png")
```

14.2 Errors

Unparseable input raises `utility_viz.ParseError`, whose message lists the supported forms:

```
from utility_viz import ParseError, parse_latex

try:
    model = parse_latex(r"x^2 + y^2")
except ParseError as e:
    print(e)

# Unrecognised LaTeX utility function: 'x^2 + y^2'
# Supported forms:
#   Cobb-Douglas       : x^{alpha} y^{beta}
#   Leontief           : \min(ax, by)
```

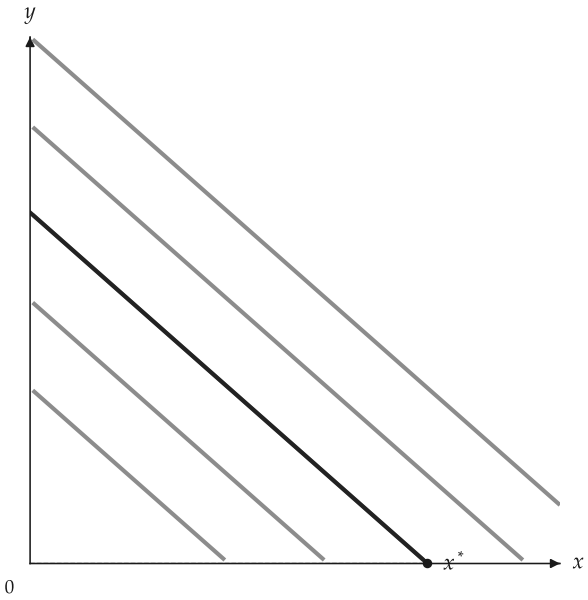


Figure 35 `parse_latex(r"2x + 3y").`

```
#   Perfect Substitutes: ax + by
#   CES                : (alpha x^{rho} + beta y^{rho})^{1/rho}
```

14.3 In the command-line tool

`utility-viz plot --latex` uses the same parser and accepts the same input (Section 11.2):

```
utility-viz plot --latex "x^{0.4} y^{0.6}" --px 2 --py 3 --income 30 -o
out.png
```

Change history

Release history of utility-viz. Documentation-only changes are not listed; each entry is tagged where the feature it describes is documented, following l3doc convention, so the page number is the real page.

v2.0.0b1 (2026-10-03)

General: Renamed from econ-viz to utility-viz: distribution, import package, command and configuration file ····· 2

Internals reorganised into `core` and `models`; the public API is the package root and `utility_viz.models` ····· 2

New dependencies mosaickit and bezierkit ····· 2

The whole repository passes `mypy`; Ruff enforces line length, complexity and statement limits ····· 2

`utility-viz`: New dependencies `mosaickit>=0.5.1,<0.6.0` and `bezierkit>=0.5.0rc1,<0.6.0` · 3

The repository is a uv workspace releasing utility-viz and econ-viz in lockstep ····· 3

The command is now `utility-viz`; `econ-viz` prints a deprecation warning and forwards to it · 51

`econ-viz`: Renamed to utility-viz; the `econ_viz` package, the `econ-viz` command and `econ-viz.toml` lookup remain as a deprecated compatibility layer until 3.0.0 ····· 4

The compatibility layer ships as a separate econ-viz distribution, so upgrading a 1.x installation no longer deletes files shared with utility-viz ····· 4

`UtilityVizError`: `UtilityVizError` is the base exception; `EconVizError` remains as an alias of the same class ····· 8

`Leontief`: Indifference curves of kinked utilities keep exact right-angle corners in Matplotlib and TikZ output ····· 30

`core.scenes`: Backend-neutral mosaickit layer factories for budget lines, equilibria and indifference curves, with native Bézier TikZ export ····· 39

`Config`: Default file renamed `utility-viz.toml`; legacy `econ-viz.toml` still read with a warning ·· 48

`Config.discover`: New `Config.discover()`, falling back to the built-in defaults when no file exists ····· 48

`utility-viz plot`: Without `--config`, reads `utility-viz.toml` (or legacy `econ-viz.toml`) from the current directory ····· 51

`utility-viz init`: New `--migrate` option ····· 51

`Animator`: Moved to `utility_viz.core.animation` ····· 55

`WidgetViewer`: Moved to `utility_viz.core.interactive` ····· 58

v1.12.0 (2026-09-26)

General: Focal and secondary indifference-curve levels, with numeric or ordinal labels ····· 2

`Canvas.add_utility`: Focal utility level, secondary curve style and ordinal labels ····· 8

`Theme`: Colour, width and opacity of secondary indifference curves ····· 42

v1.11.0 (2026-09-26)

General: Complete theme system: one visual language across canvases, multi-panel figures, demand diagrams, decompositions and Edgeworth boxes ····· 2

`Canvas`: Backgrounds, labels, lines and markers all take their defaults from the theme ····· 8

`Figure`: Panel backgrounds, labels, lines and markers fully follow the theme ····· 17

`EdgeworthBox`: Box frame, contract curve, core, endowment, price line and Walrasian equilibrium fully follow the theme ····· 26

`Theme`: Complete theme system: background colour, label scale and defaults for every economic layer ····· 42

`themes`: New `paper`, `monochrome`, `presentation` and `dark` themes ····· 42

`Config`: Configuration files accept every built-in theme and keep the same visual meaning across diagram types ····· 48

`econ-viz plot`: `--theme` accepts every built-in CLI theme ····· 51

v1.10.1 (2026-09-26)

General: Ships `py.typed`; CI runs Ruff and Mypy ····· 2

v1.10.0 (2026-09-26)

`econ-viz`: Python 3.10 reads TOML configuration files with tomli ····· 3

`Figure`: Falls back to the active `Config` when no theme or font is given ····· 17

`EdgeworthBox`: Falls back to the active `Config` when no theme or font is given ····· 26

`Config`: `Config`, `econ-viz.toml`, `econ-viz init` and `plot --config` ····· 48

`econ-viz init`: Configuration-template command and `plot --config` ····· 51

v1.9.0 (2026-09-26)

`Canvas`: Axes, origin, titles and other text elements accept a `Label` ····· 8

`Stroke`: New `opacity` field ····· 14

`DemandDiagram`: Supports `Legend`, `Marker`, `Label` and per-layer `Stroke` ····· 19
`Canvas.add_decomposition`: Draws the indifference curves through A, B and C by default, with curve labels and `Legend` ····· 22
Effect arrows below the horizontal axis point A→B and B→C; zero effects draw no arrow ···· 22
`Haagsma`: Haagsma utility model with a closed-form solution, for inferior- and Giffen-good analysis ····· 34
v1.8.0 (2026-09-26)
`Canvas`: `Axis`, `Marker`, `Label` and `Fill` style objects ····· 8
`Stroke`: Every line style is a `Stroke`; the individual style arguments remain as shorthands ····· 14
`Canvas.add_decomposition`: `Effect`, `Marker` and `Label` style objects ····· 22
`EdgeworthBox`: Markers, text and axes accept `Marker`, `Label` and `Axis` ····· 26
v1.7.0 (2026-09-25)
General: CI tests Python 3.10–3.13, enforces branch coverage, and smoke-tests runnable examples ····· 2
Tagged releases publish only after tests pass ···· 2
`econ-viz`: Package management and builds migrated to uv ····· 3
All example scripts run from a clean checkout · 4
`econ-viz --version` prints the installed package version ····· 51
`Canvas`: Per-figure text and math fonts without changing matplotlib's global configuration ····· 8
Independent axis-label positions, line styles and arrowhead styles for `Canvas` and `Figure` ····· 14
`Stroke`: One `Stroke` controls line width, style, colour and arrowheads across canvases, figures, demand diagrams and Edgeworth boxes ····· 14
`CobbDouglas`: Fixed utility domains, asymmetric CES expansion paths, Cobb-Douglas limit cases, and satiation optima that leave income unspent · 28
Fixed contour levels when utility is near zero or negative ····· 28
`comparative_statics`: Comparative statics no longer step outside the domain near zero prices or income, or near subsistence constraints ····· 40
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General: Generated files under `examples/output/` are no longer tracked ····· 2
`Canvas`: Default indifference-curve line width reduced from `2.0` to `1.8` ····· 14
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`Canvas.save`: TikZ axis regression tests no longer depend on colour indices ····· 51
`Animator`: Animation examples moved to `examples/scripts/animation.py` ····· 55
v1.5.0 (2026-04-21)
General: New example scripts for decomposition, demand diagrams, PCC/ICC paths, multi-panel layouts and TikZ export ····· 2
Regression tests for TikZ export and price-effect decomposition ····· 2
`econ-viz`: Notebook install flow on Colab is restart-safe ····· 4
`Figure`: Better placement and visibility of axis labels in shared-panel layouts ····· 17
`decompose_price_effect`: Price-effect decomposition: `decompose_price_effect()`, `PriceEffectDecomposition` and `Canvas.add_decomposition()`, drawing bundles A/B/C with substitution and income effects ····· 22
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v1.4.0 (2026-04-09)
General: Tests for `Animator` and `WidgetViewer` · 2
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